

Chain Conditions in Products

The chain conditions introduced in Chapter 2 are studied here in cartesian products and powers, both with the usual cartesian product topology and with various strong κ -box topologies. Most of the results follow this pattern: if every ‘small’ subproduct (or, every ‘small’ power of a space) satisfies a particular chain condition, then the full ‘large’ product (or, every power of the space) also satisfies it. The combinatorial tool suitable for this purpose is the Erdős–Rado theorem on quasi-disjoint families and its extension to singular cardinals.

Our results are more complete and satisfactory for the usual product topology than for the κ -box topologies, and for chain conditions concerning calibre and compact-calibre than for pseudo-compactness numbers. The distinction we draw between powers and products is not artificial, since some chain conditions are preserved in the former context and not in the latter (compare, for example, Theorem 3.18(a) (of Shelah) or its extension Theorem 3.19(a) (based on an argument of Argyros) with Theorem 4.4 (of Gerlits), and compare Theorem 3.18(b) or Theorem 3.19(b) with Theorem 8.3).

The chart in section 3.24 summarizes what we know about chain conditions in products and powers, and indicates some open questions.

Notation. Let $\{X_i : i \in I\}$ be a family of sets and $X = \prod_{i \in I} X_i$. If $J \subset I$ and $J \neq \emptyset$, we set

$$X_J = \prod_{i \in J} X_i,$$

and we denote by π_J the projection function from X onto X_J given by $\pi_J(\langle p_i : i \in I \rangle) = \langle p_i : i \in J \rangle$. In particular, $X = X_I$. If $p \in X$, we

sometimes write p_j for $\pi_j(p)$, though we write (as usual) π_i and p_i for $\pi_{(i)}$ and $p_{(i)}$, respectively.

A subset A of X such that

$$A = \prod_{i \in I} A_i,$$

where $A_i \subset X_i$ for $i \in I$, is called a (*generalized*) *rectangle* in X , and the set

$$R(A) = \{i \in I : A_i \neq X_i\}$$

is called the *restriction set* of A .

3.1 Lemma. Let $A = \prod_{i \in I} A_i$ and $B = \prod_{i \in I} B_i$ be non-empty generalized rectangles in the product $X = \prod_{i \in I} X_i$, and let $R(A) \cap R(B) \subset J \subset I$. Then the following are equivalent.

- (a) $A \cap B = \emptyset$;
- (b) there is $i \in J$ such that $A_i \cap B_i = \emptyset$;
- (c) $J \neq \emptyset$ and $\pi_J[A] \cap \pi_J[B] = \emptyset$.

Proof. (a) $\Rightarrow$ (b). If (b) fails there is $x = \langle x_i : i \in I \rangle \in X$ such that

$$\begin{aligned} x_i &\in A_i \cap B_i && \text{if } i \in J \\ x_i &\in A_i && \text{if } i \in R(A) \setminus J \\ x_i &\in B_i && \text{if } i \in R(B) \setminus J \\ x_i &\in X_i && \text{if } i \in I \setminus (R(A) \cup R(B)). \end{aligned}$$

Then $x \in A \cap B$ and hence (a) fails.

(b) $\Rightarrow$ (c) is obvious.

(c) $\Rightarrow$ (a). If $x \in A \cap B$ then $x_J \in \pi_J[A] \cap \pi_J[B]$.

Definition. Let $\{X_i : i \in I\}$ be a family of spaces, let X be the set $\prod_{i \in I} X_i$ and let κ be an infinite cardinal. The κ -box topology on X is the topology with base (called the *canonical base*) equal to the family of all subsets U of X such that

$$\begin{aligned} U &= \prod_{i \in I} U_i, \\ U_i &\text{ is open in } X_i \text{ for } i \in I, \text{ and} \\ |R(U)| &< \kappa. \end{aligned}$$

It is simple to verify that the canonical base for the κ -box topology just defined is indeed a base (for a topology). The product set

$X_I = \prod_{i \in I} X_i$ with the κ -box topology is denoted by $(X_I)_\kappa$ or $(\prod_{i \in I} X_i)_\kappa$.

The ω -product topology is the usual product topology; we write simply X_I and $\prod_{i \in I} X_i$ for $(X_I)_\omega$ and $(\prod_{i \in I} X_i)_\omega$, respectively.

In discussing the κ -box topology for a product and its subsets we use, among the various bases for it that exist, only the canonical base. Thus for $\{X_i : i \in I\}$ and κ as above and $Y \subset \prod_{i \in I} X_i$ the expression ‘ V is basic in Y ’ means that there is an element U of the canonical base for $(\prod_{i \in I} X_i)_\kappa$ such that $V = U \cap Y$.

In the next lemma and elsewhere in this monograph we allow ourselves the following convenient abuse of notation: Given $\{X_i : i \in I\}$, $\emptyset \neq J \subsetneq I$ and $\kappa \geq \omega$, we identify X_I with $X_J \times X_{I \setminus J}$ and $(X_I)_\kappa$ with $(X_J)_\kappa \times (X_{I \setminus J})_\kappa$.

3.2 Lemma. Let $\kappa \geq \omega$, let $\{X_i : i \in I\}$ be a set of spaces, and $J \in \mathcal{P}^*(I)$.

(a) The projection function $\pi_J : (X_I)_\kappa \rightarrow (X_J)_\kappa$ is an open, continuous function;

(b) if K is a compact subset of $(X_J)_\kappa$ and $p \in X_{I \setminus J}$, then $\{p\} \times K$ is a compact subset of $(X_I)_\kappa$.

We omit the simple proof of this lemma.

In sections 3.3–3.7 we show that for certain of the properties P we have been considering, the question whether a product $(X_I)_\kappa$ (or even certain of its dense subspaces) has P is determined by whether certain of the subproducts $(X_J)_\kappa$ have P .

In the interest of completeness we give first a particularly simple result. Its proof does not depend on the Erdős–Rado Theorem.

We note that in Theorems 3.3, 3.6, 3.9, 3.12, 3.13 and 3.25 the space $Y = (X_I)_\kappa$ satisfies the conditions imposed on Y . This is the case of principal interest.

3.3 Theorem. Let $\alpha \geq \beta \geq \gamma \geq 2$, $\alpha \geq \kappa \geq \omega$, let $\{X_i : i \in I\}$ be a set of spaces and Y a subspace of $(X_I)_\kappa$.

(a) If $(X_J)_\kappa$ has calibre (α, β, γ) for every non-empty $J \subset I$ with $|J| \leq \alpha$, and if $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| \leq \alpha$, then Y has calibre (α, β, γ) ;

(b) if $(X_J)_\kappa$ has compact-calibre (α, β, γ) for every non-empty

$J \subset I$ with $|J| \leq \alpha$, and if for every compact $K_J \subset (X_J)_\kappa$ with J non-empty, $J \subset I$ and $|J| \leq \alpha$ there is compact $K \subset Y$ such that $\pi_J[K] = K_J$, then Y has compact-calibre (α, β, γ) ;

(c) if $(X_J)_\kappa$ is pseudo- (α, β) -compact for every non-empty $J \subset I$ with $|J| \leq \alpha$, and if $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| \leq \alpha$, then Y is pseudo- (α, β) -compact.

Proof. We assume in what follows, the statements being obvious otherwise, that $X_i \neq \emptyset$ for $i \in I$.

Let $\{U^\xi : \xi < \alpha\}$ be a set of non-empty basic subsets of $(X_I)_\kappa$ and set $J = \cup_{\xi < \alpha} R(U^\xi)$. We note that $|J| \leq \alpha$.

(a) There is $B \in [\alpha]^\beta$ such that if $C \in [B]^\gamma$ then $\cap_{\xi \in C} \pi_J[U^\xi] \neq \emptyset$. Let $C \in [B]^\gamma$ and $p \in \cap_{\xi \in C} \pi_J[U^\xi]$ and let x be an element of Y such that $\pi_J(x) = p$. It is clear that $x \in \cap_{\xi \in C} U^\xi$.

(b) There is $B \in [\alpha]^\beta$ such that for $C \in [B]^\gamma$ there is compact $K_J \subset (X_J)_\kappa$ such that $\pi_J[U^\xi] \cap K_J \neq \emptyset$ whenever $\xi \in C$. Let $C \in [B]^\gamma$, let K_J be such a compact subspace of $(X_J)_\kappa$ and let K be a compact subspace of Y such that $\pi_J[K] = K_J$. It is clear that $U^\xi \cap K \neq \emptyset$ whenever $\xi \in C$.

(c) There is $p \in (X_J)_\kappa$ such that

$$|\{\xi < \alpha : V_J \cap \pi_J[U^\xi] \neq \emptyset\}| \geq \beta$$

for every neighborhood V_J of p . It is clear that if x is an element of Y such that $\pi_J(x) = p$, then

$$|\{\xi < \alpha : V \cap U^\xi \neq \emptyset\}| \geq \beta$$

for every neighborhood V of x .

3.4 Corollary. Let $\{X_i : i \in I\}$ be a set of spaces such that X_J is pseudocompact for every non-empty $J \subset I$ with $|J| \leq \omega$. Then X_I is pseudocompact.

Proof. This is the case $\alpha = \beta = \kappa = \omega$, $Y = X_I$ of Theorem 3.3(c).

We show next that for $\kappa \ll \text{cf}(\alpha)$ (and without additional hypotheses) the behavior of products of fewer than $\text{cf}(\alpha)$ spaces with respect to the chain conditions we are considering is determined by the behavior of subproducts of fewer than κ spaces.

3.5 Theorem. Let $\alpha \geq \beta \geq \gamma \geq 2$ and $\omega \leq \kappa \ll \text{cf}(\alpha)$ and let $\{X_i : i \in I\}$ be a set of spaces.

(a) If $(X_J)_\kappa$ has calibre (α, β, γ) for every non-empty $J \subset I$ with $|J| < \kappa$, then $(X_J)_\kappa$ has calibre (α, β, γ) for every non-empty $J \subset I$ with $|J| < \text{cf}(\alpha)$;

(b) if $(X_J)_\kappa$ has compact-calibre (α, β, γ) for every non-empty $J \subset I$ with $|J| < \kappa$, then $(X_J)_\kappa$ has compact-calibre (α, β, γ) for every non-empty $J \subset I$ with $|J| < \text{cf}(\alpha)$;

(c) if $(X_J)_\kappa$ is pseudo- (α, β) -compact for every non-empty $J \subset I$ with $|J| < \kappa$, then $(X_J)_\kappa$ is pseudo- (α, β) -compact for every non-empty $J \subset I$ with $|J| < \text{cf}(\alpha)$.

Proof. We assume in what follows, the statements being obvious otherwise, that $X_i \neq \emptyset$ for $i \in I$. We assume without loss of generality that $|I| < \text{cf}(\alpha)$ and we show in each case that $(X_I)_\kappa$ has the property in question.

Let $\{U^\xi: \xi < \alpha\}$ be a set of non-empty, basic subsets of $(X_I)_\kappa$. Since

$$R(U^\xi) \in \mathcal{P}_\kappa(I) \text{ for } \xi < \alpha \text{ and} \\ |\mathcal{P}_\kappa(I)| = \sum_{\lambda < \kappa} |I|^\lambda < \text{cf}(\alpha),$$

there are $A \in [\alpha]^\alpha$ and $J \subset I$ such that

$$R(U^\xi) = J \text{ for } \xi \in A.$$

If $J = \emptyset$ then $U^\xi = (X_I)_\kappa$ for all $\xi \in A$ and hence $\bigcap_{\xi \in A} U^\xi \neq \emptyset$, and if $J = I$ then $|I| < \kappa$; in either case the required relations follow immediately. We assume

$$\emptyset \neq J \subset I$$

and we choose $q \in X_{I \setminus J}$.

In (a) there is $B \in [A]^\beta$ such that if $C \in [B]^\gamma$ there is

$$p(C) \in \bigcap_{\xi \in S} \pi_J[U^\xi];$$

it follows with $x(C) = \langle p(C), q \rangle \in X_J$ that $x(C) \in \bigcap_{\xi \in C} U^\xi$.

In (b) there is $B \in [A]^\beta$ such that if $C \in [B]^\gamma$ there is compact $K(C) \subset (X_J)_\kappa$ such that $\pi_J[U^\xi] \cap K(C) \neq \emptyset$ whenever $\xi \in C$. Then $K(C) \times \{q\}$ is a compact subspace of $(X_I)_\kappa$ and

$$U^\xi \cap (K(C) \times \{q\}) \neq \emptyset \text{ whenever } \xi \in C.$$

In (c) there is $p \in (X_J)_\kappa$ such that

$$|\{\xi \in A : V_J \cap \pi_J[U^\xi] \neq \emptyset\}| \geq \beta$$

for every neighborhood V_J of p . It follows with

$$x = \langle p, q \rangle \in (X_I)_\kappa$$

that

$$|\{\xi \in A : V \cap U^\xi \neq \emptyset\}| \geq \beta$$

for every neighborhood V of x .

The proof is complete.

Product theorems for regular cardinals

3.6 Theorem. Let $\alpha \geq \beta \geq \gamma \geq 2$, $\omega \leq \kappa \ll \alpha$ with α regular, and let $\{X_i : i \in I\}$ be a set of spaces and $Y \subset (X_I)_\kappa$.

(a) If $(X_J)_\kappa$ has calibre (α, β, γ) for every non-empty $J \subset I$ with $|J| < \kappa$, and if $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| \leq \gamma \cdot \kappa$, then Y has calibre (α, β, γ) ;

(b) if $(X_J)_\kappa$ has compact-calibre (α, β, γ) for every non-empty $J \subset I$ with $|J| < \kappa$, and if for every compact $K_J \subset (X_J)_\kappa$ with J non-empty, $J \subset I$ and $|J| \leq \gamma \cdot \kappa$ there is compact $K \subset Y$ such that $\pi_J[K] = K_J$, then Y has compact-calibre (α, β, γ) ;

(c) if $(X_J)_\kappa$ is pseudo- (α, β) -compact for every non-empty $J \subset I$ with $|J| < \kappa$, and if $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| < \kappa$, and if $\beta \geq \kappa$, then Y is pseudo- (α, β) -compact.

Proof. We assume in what follows, the statements being obvious otherwise, that $X_i \neq \emptyset$ for $i \in I$.

Let $\{U^\xi : \xi < \alpha\}$ be a set of non-empty basic subsets of $(X_I)_\kappa$. Since $|R(U^\xi)| < \kappa$ for $\xi < \alpha$, by the Erdős–Rado theorem on quasi-disjoint families (Theorem 1.4) there are $A \in [\alpha]^\alpha$ and J such that if $\xi, \xi' \in A$ and $\xi \neq \xi'$ then $R(U^\xi) \cap R(U^{\xi'}) = J$. We consider statements (a), (b) and (c) separately.

(a) We show there is $B \in [A]^\beta$ such that if $C \in [B]^\gamma$ then $(\cap_{\xi \in C} U^\xi) \cap Y \neq \emptyset$.

If $J = \emptyset$ we let B be an arbitrary element of $[A]^\beta$, and if $J \neq \emptyset$ we choose $B \in [A]^\beta$ such that if $C \in [B]^\gamma$ then $\cap_{\xi \in C} \pi_J[U^\xi] \neq \emptyset$.

Now let $C \in [B]^\gamma$, set $F = \cup_{\xi \in C} R(U^\xi)$, and if $J \neq \emptyset$ let $p \in \cap_{\xi \in C}$

$\pi_J[U^\xi]$. Since $|F \cup J| \leq \gamma \cdot \kappa + \kappa = \gamma \cdot \kappa$ we have $\pi_F[Y] = X_F$; it follows that since

$$|\{\xi \in C : i \in R(U^\xi)\}| \leq 1 \text{ for } i \in F \setminus J,$$

there is $x = \langle x_i : i \in I \rangle \in Y$ such that

$$\begin{aligned} x_i &= p_i \text{ if } i \in J \\ &\in U_i^\xi \text{ if } i \in F \setminus J \text{ with } i \in R(U^\xi), \xi \in C, \\ &\in X_i \text{ if } i \in I \setminus F. \end{aligned}$$

It is clear that $x \in (\bigcap_{\xi \in C} U^\xi) \cap Y$.

(b) We show there is $B \in [A]^\beta$ such that for $C \in [B]^\gamma$ there is compact $K \subset Y$ such that $U^\xi \cap K \neq \emptyset$ whenever $\xi \in C$.

If $J = \emptyset$ we let B be an arbitrary element of $[A]^\beta$, and if $J \neq \emptyset$ we choose $B \in [A]^\beta$ such that for $C \in [B]^\gamma$ there is compact $\tilde{K} \subset (X_J)_\kappa$ such that $\pi_J[U^\xi] \cap \tilde{K} \neq \emptyset$ whenever $\xi \in C$.

Now let $C \in [B]^\xi$, set $F = \bigcup_{\xi \in C} R(U^\xi)$, and if $J \neq \emptyset$ choose compact $\tilde{K} \subset (X_J)_\kappa$ such that $\pi_J[U^\xi] \cap \tilde{K} \neq \emptyset$ whenever $\xi \in C$. Since

$$|\{\xi \in C : i \in R(U^\xi)\}| \leq 1 \text{ for } i \in F \setminus J,$$

there is $q \in X_{F \cup J}$ such that

$$q_i \in U_i^\xi \text{ if } i \in F \setminus J, i \in R(U^\xi), \xi \in C.$$

We set $K_{F \cup J} = \{q\} \times \tilde{K}$. Then $K_{F \cup J}$ is by Lemma 3.2(b) a compact subspace of $(X_{F \cup J})_\kappa$, and since

$$|F \cup J| \leq \gamma \cdot \kappa + \kappa = \gamma \cdot \kappa,$$

there is compact $K \subset Y$ such that $\pi_{F \cup J}[K] = K_{F \cup J}$. It is clear that $U^\xi \cap K \neq \emptyset$ whenever $\xi \in C$, as required.

(c) We show there is $x \in Y$ such that if V is a neighborhood in $(X_J)_\kappa$ of x then

$$|\{\xi \in A : V \cap U^\xi \neq \emptyset\}| \geq \beta;$$

this is sufficient, since from $V \cap U^\xi \neq \emptyset$ we have $V \cap U^\xi \cap Y \neq \emptyset$ (because Y is dense in $(X_J)_\kappa$).

If $J = \emptyset$ we let x be an arbitrary element of Y , and if $J \neq \emptyset$ we use the fact that $(X_J)_\kappa$ is pseudo- (α, β) -compact and $\pi_J[Y] = X_J$ to find $x \in Y$ such that

$$|\{\xi \in A : V_J \cap \pi_J[U^\xi] \neq \emptyset\}| \geq \beta$$

for every neighborhood V_J of x_J in $(X_J)_\kappa$.

Now let V be a basic neighborhood in $(X_I)_\kappa$ of x , and set $V_J = \pi_J[V]$ and

$$B = \{\xi \in A : V_J \cap \pi_J[U^\xi] \neq \emptyset\}.$$

If $\xi \in B$ and $V \cap U^\xi = \emptyset$, then from Lemma 3.1 we have $(R(V) \cap R(U^\xi)) \setminus J \neq \emptyset$. Since

$$|\{\xi \in B : i \in R(U^\xi) \setminus J\}| \leq 1 \text{ for } i \in I,$$

we have

$$|\{\xi \in B : V \cap U^\xi = \emptyset\}| \leq |R(V)| < \kappa \leq \beta,$$

and from $|B| \geq \beta$ we have

$$|\{\xi \in B : V \cap U^\xi \neq \emptyset\}| \geq \beta,$$

as required.

The proof is complete.

In Theorem 3.6(c) (and Theorem 3.12(c) below) the condition $\beta \geq \kappa$ is superfluous in case $\kappa = \omega$ and the spaces X_i are regular, Hausdorff spaces (cf. in this connection Theorem 2.1(i)).

We recall that a space X is said to have precalibre (α, β) if for every set $\{U_\xi : \xi < \alpha\}$ of non-empty, open subsets of X there is $B \subset \alpha$ with $|B| = \beta$ such that $\{U_\xi : \xi \in B\}$ has the finite intersection property. We note now that the argument in the proof of Theorem 3.6(a) shows the following statement.

Theorem. Let $\alpha \geq \beta \geq \omega, \omega \leq \kappa \ll \alpha$ with α regular and let $\{X_i : i \in I\}$ be a set of spaces such that $(X_J)_\kappa$ has precalibre (α, β) for every non-empty $J \subset I$ with $|J| < \kappa$. Then $(X_I)_\kappa$ has precalibre (α, β) .

3.7 Corollary. Let α be an uncountable, regular cardinal.

- (a) The product of every set of spaces with calibre α has calibre α ;
- (b) the product of every set of spaces with compact-calibre α has compact-calibre α .

Proof. This follows from Theorem 2.2 and (the case $\alpha = \beta = \gamma, \kappa = \omega, Y = (X_I)_\kappa$ of) Theorem 3.6.

3.8 Remark. Let α, β and κ be cardinals with $\alpha \geq \beta$. The notation

$$\alpha \rightarrow \Delta(\kappa, \beta)$$

means that if $\{S_\xi : \xi < \alpha\}$ is a family of sets such that $|S_\xi| < \kappa$ for all $\xi < \alpha$, then there are $A \in [\alpha]^\beta$ and J such that

$$S_\xi \cap S_{\xi'} = J \text{ for } \xi, \xi' \in A, \xi \neq \xi'.$$

In this notation the Erdős–Rado theorem on quasi-disjoint sets (Theorem 1.4) is the statement that if $\omega \leq \kappa \ll \alpha$ with α regular then $\alpha \rightarrow \Delta(\kappa, \alpha)$. Since our concern in this work (as to cardinals) is chiefly with pairs $\langle \alpha, \kappa \rangle$ such that $\omega \leq \kappa \ll \alpha$ and with the extension to singular cardinals of the Erdős–Rado theorem, we do not consider the general relation $\alpha \rightarrow \Delta(\kappa, \beta)$. Nevertheless we note here in passing that the proof of Theorem 3.6 becomes, with only minor modifications (not given in detail here), a proof of the following statement.

Theorem. Let $\alpha \geq \beta \geq \gamma \geq \delta \geq 2$, $\kappa \geq \omega$, $\alpha \rightarrow \Delta(\kappa, \beta)$, let $\{X_i : i \in I\}$ be a set of spaces and $Y \subset (X_i)_\kappa$.

(a) If $(X_i)_\kappa$ has calibre (β, γ, δ) for every non-empty $J \subset I$ with $|J| < \kappa$, and if $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| \leq \delta \cdot \kappa$, then Y has calibre (α, γ, δ) ;

(b) if $(X_i)_\kappa$ has compact-calibre (β, γ, δ) for every non-empty $J \subset I$ with $|J| < \kappa$, and if for every compact $K_J \subset (X_J)_\kappa$ with J non-empty, $J \subset I$ and $|J| \leq \delta \cdot \kappa$ there is compact $K \subset Y$ such that $\pi_J[K] = K_J$, then Y has compact-calibre (α, γ, δ) ;

(c) if $(X_i)_\kappa$ is pseudo- (β, γ) -compact for every non-empty $J \subset I$ with $|J| < \kappa$ and if $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| < \kappa$, and if $\gamma \geq \kappa$, then Y is pseudo- (α, γ) -compact.

The methods of 3.6–3.8 are not applicable to the cardinal number $\alpha = \omega$. Indeed we have noted already in Theorem 2.10 that no infinite Hausdorff space has calibre ω , and we show below (the case $\alpha = \beta = \omega$ of Theorem 9.7(a)) that the product of a finite number of completely regular, Hausdorff, pseudocompact spaces need not be pseudocompact. We show now, in contrast, by a special argument not dependent on the Erdős–Rado theorem on quasi-disjoint sets, that the property compact-calibre ω is productive.

3.9 Theorem. Let $\{X_i : i \in I\}$ be a set of non-empty spaces, each with compact-calibre ω , and let $Y \subset X_I$. If for every compact $K_J \subset$

X_J with J non-empty, $J \subset I$ and $|J| \leq \omega$ there is compact $K \subset Y$ such that $\pi_J[K] = K_J$, then Y has compact-calibre ω .

Proof. Let $\{U^n : n < \omega\}$ be a set of non-empty basic subsets of X_I . It is enough to show there are $A \in [\omega]^\omega$ and compact $K \subset Y$ such that $U^n \cap K \neq \emptyset$ whenever $n \in A$. We set $J = \cup_{n < \omega} R(U^n)$, we assume without loss of generality that $J \neq \emptyset$, we note that $|J| \leq \omega$ since $|R(U^n)| < \omega$ for $n < \omega$, and we consider two cases.

Case 1. $|J| < \omega$. From Theorem 2.2(b) there are $A \in [\omega]^\omega$ and compact $K_J \subset X_J$ such that $\pi_J[U^n] \cap K_J \neq \emptyset$ whenever $n \in A$. We choose compact $K \subset Y$ such that $\pi_J[K] = K_J$; it is clear that A and K are as required.

Case 2. $|J| = \omega$. We write $J = \{i(k) : k < \omega\}$ with $i(k) \neq i(k')$ when $k \neq k'$, and we define

$$\{A_k : k < \omega\} \subset \mathcal{P}(\omega) \text{ and } \{K_k : k < \omega\} \text{ so that}$$

- (i) $A_0 = \omega$;
- (ii) $A_{k+1} \in [A_k]^\omega$ for $k < \omega$;
- (iii) K_k is a compact subspace of $X_{i(k)}$ for $k < \omega$; and
- (iv) $U_{i(k)}^n \cap K_k \neq \emptyset$ whenever $n \in A_{k+1}$.

We proceed by recursion. We define A_0 by (i), and for $k < \omega$ we note that since $\{U_{i(k)}^n : n \in A_k\} \subset \mathcal{T}^*(X_{i(k)})$ and $X_{i(k)}$ has compact-calibre ω , there are A_{k+1} and K_k satisfying (ii), (iii) and (iv).

Since $A_{k+1} \in [A_k]^\omega$ for $k < \omega$, there is $\{n(m) : m < \omega\}$ such that

$$\begin{aligned} n(m) &\in A_m \text{ for } m < \omega, \text{ and} \\ n(m') &< n(m) \text{ for } m' < m < \omega. \end{aligned}$$

We set $A = \{n(m) : m < \omega\}$, we choose

$$x_{k,m} \in U_{i(k)}^{n(m)} \text{ for } k, m < \omega,$$

we set

$$\begin{aligned} \tilde{K}_k &= K_k \cup \{x_{k,m} : m \leq k\} \text{ for } k < \omega \text{ and} \\ K_J &= \prod_{k < \omega} \tilde{K}_k, \end{aligned}$$

we note that K_J is a compact subset of X_J and we choose compact $K \subset Y$ such that $\pi_J[K] = K_J$.

To show that $U^{n(m)} \cap K \neq \emptyset$ for $m < \omega$ it is sufficient, since $R(U^{n(m)}) \subset J$, to show that

$$\pi_J[U^{n(m)}] \cap K_J \neq \emptyset,$$

i.e.,

$$U_i^{n(m)} \cap \pi_i[K] \neq \emptyset \text{ for } i \in J.$$

If $i = i(k) \in J$ with $k < m$ then from $n(m) \in A_m \subset A_{k+1}$ we have

$$U_i^{n(m)} \cap \pi_i[K] \supset U_i^{n(m)} \cap K_k \neq \emptyset;$$

and if $i = i(k) \in J$ with $k \geq m$ then

$$x_{k,m} \in U_{i(k)}^{n(m)} \cap \tilde{K}_k = U_i^{n(m)} \cap \pi_i[K].$$

The proof is complete.

Product theorems for singular cardinals

We show first that the statements of the various parts of Theorem 3.6 are all false for singular cardinals $\alpha > \beta$; the correct analogue of Theorem 3.6 (requiring additional hypotheses) is given in Theorem 3.13 and its corollaries.

3.10 Theorem. Let α be an (infinite) singular cardinal. There is a set $\{X_i : i \in I\}$ of completely regular, Hausdorff spaces such that

(i) if $\alpha > \beta \geq 2$, $\omega \leq \kappa \leq \alpha$ and $\kappa \leq \text{cf}(\alpha)$, then $(X_J)_\kappa$ has calibre (α, β) for all $J \subset I$ with $|J| < \kappa$; and

(ii) if $\omega \leq \kappa$ then $(X_J)_\kappa$ is not pseudo- $(\alpha, 2)$ -compact.

Proof. There is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ of cardinal numbers such that

$$\alpha_0 = \text{cf}(\alpha),$$

$$\alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and}$$

$$\sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma = \alpha.$$

We set $I = \text{cf}(\alpha)$ and $X_\sigma = \alpha_\sigma$ for $\sigma \in I$. We verify (i) and (ii).

(i) Since $|J| < \kappa$, the space $(X_J)_\kappa$ is discrete. We set $\gamma = |(X_J)_\kappa|$ and we note from Theorem A.5(a) of the Appendix that $\gamma < \alpha$. It follows from Theorem 2.4(c) that the space $(X_J)_\kappa$, which is the discrete space γ , has calibre (α, β) for all β such that $\alpha > \beta \geq \omega$. In particular $(X_J)_\kappa$ has calibre (α, ω) and hence calibre (α, β) for $\beta < \omega$.

(ii) It is enough to show that there is a (faithfully indexed) cellular family $\{U(s) : s \in S\}$ in X_I with $|S| = \alpha$ such that

$$X_I = (X_I)_\omega = \bigcup_{s \in S} U(s).$$

We set

$$U(0) = \{x \in X_I : x_0 = 0\}, \text{ and}$$

$$U(\sigma, \eta) = \{x \in X_I : x_0 = \sigma, x_\sigma = \eta\} \text{ for } 0 < \sigma < \text{cf}(\alpha), \eta < \alpha_\sigma.$$

It is clear that the family

$$\{U(\sigma, \eta) : 0 < \sigma < \text{cf}(\alpha), \eta < \alpha_\sigma\} \cup \{U(0)\},$$

indexed by the set $\cup \{\{\sigma\} \times \alpha_\sigma : 0 < \sigma < \text{cf}(\alpha)\} \cup \{0\}$, is as required.

3.11 Corollary. Let α be an (infinite) singular cardinal, $\alpha > \beta \geq 2$, $\omega \leq \kappa \ll \alpha$ and $\kappa \leq \text{cf}(\alpha)$. There is a set $\{X_i : i \in I\}$ of completely regular, Hausdorff spaces such that

- (i) $(X_J)_\kappa$ has calibre (α, β) for every non-empty $J \subset I$ with $|J| < \kappa$, and $(X_I)_\kappa$ does not have calibre (α, β) ;
- (ii) $(X_J)_\kappa$ has compact-calibre (α, β) for every non-empty $J \subset I$ with $|J| < \kappa$, and $(X_I)_\kappa$ does not have compact-calibre (α, β) if $\beta \geq \omega$; and
- (iii) $(X_J)_\kappa$ is pseudo- (α, β) -compact for every non-empty $J \subset I$ with $|J| < \kappa$, and $(X_I)_\kappa$ is not pseudo- (α, β) -compact.

Proof. It follows from Theorem 2.1 that the set $\{X_i : i \in I\}$ of Theorem 3.10 has the required properties.

We remark that it follows from Theorem 3.5 that among all sets I which may index a set of spaces with the properties of those in 3.10 and 3.11, the present set I (satisfying $|I| = \text{cf}(\alpha)$) is of minimal cardinality.

We note that for $J \subset I$ with $0 < |J| < \kappa$ the (discrete) space $(X_J)_\kappa$ of Theorem 3.10 satisfies $|(X_J)_\kappa| \geq \text{cf}(\alpha)$; it follows from Theorem 2.4 that this space does not have calibre α , does not have compact-calibre α , and is not pseudo- (α, α) -compact.

The questions whether, with α a singular cardinal and with $\alpha = \beta$, calibre (α, β) and compact-calibre (α, β) are determined in product spaces by 'small' subproducts, are settled (in the negative, by counter-examples) in Theorems 4.4 and 8.3, respectively. Unexpectedly, the analogous question for pseudo- (α, β) -compactness has a positive answer (cf. Theorem 3.16 below).

In the interest of completeness we give first a particularly simple positive result for singular cardinals, not needed later, whose proof parallels that of Theorem 3.3.

3.12 Theorem. Let α be a singular cardinal with $\text{cf}(\alpha) > \omega$, let $\omega \leq \kappa \ll \alpha$, $\kappa \ll \text{cf}(\alpha)$ and $\alpha \geq \beta \geq \gamma \geq 2$, let $\{X_i : i \in I\}$ be a set of spaces and $Y \subset (X_I)_\kappa$.

(a) If $(X_J)_\kappa$ has calibre (α, β, γ) for all non-empty $J \subset I$ with $|J| \leq \text{cf}(\alpha)$, and if $\pi_J[Y] = X_J$ for all non-empty $J \subset I$ with $|J| \leq \gamma \cdot \text{cf}(\alpha)$, then Y has calibre (α, β, γ) ;

(b) if $(X_J)_\kappa$ has compact-calibre (α, β, γ) for all non-empty $J \subset I$ with $|J| \leq \text{cf}(\alpha)$, and if for every compact $K_J \subset (X_J)_\kappa$ with J non-empty, $J \subset I$ and $|J| \leq \gamma \cdot \text{cf}(\alpha)$ there is compact $K \subset Y$ such that $\pi_J[K] = K_J$, then Y has compact-calibre (α, β, γ) ;

(c) if $(X_J)_\kappa$ is pseudo- (α, β) -compact for every non-empty $J \subset I$ with $|J| \leq \text{cf}(\alpha)$, and if $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| \leq \text{cf}(\alpha)$, and if $\beta \geq \kappa$, then Y is pseudo- (α, β) -compact.

Proof. We assume in what follows, the statements being otherwise obvious, that $X_i \neq \emptyset$ for $i \in I$.

It follows from Theorem A.7 that there is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ of cardinal numbers such that

$$\begin{aligned} \alpha_\sigma &\text{ is regular for } \sigma < \text{cf}(\alpha), \\ \kappa &\ll \alpha_\sigma \text{ for } \sigma < \text{cf}(\alpha), \\ \text{cf}(\alpha) &< \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma &= \alpha. \end{aligned}$$

Let $\{U^\xi : \xi < \alpha\}$ be a set of non-empty basic subsets of $(X_I)_\kappa$. It follows from the extension of the Erdős–Rado theorem to singular cardinals (Theorem 1.10) that there are $\Gamma, J, \{A_\sigma : \sigma \in \Gamma\}$ and $\{J_\sigma : \sigma \in \Gamma\}$ such that

$$\begin{aligned} \Gamma &\in [\text{cf}(\alpha)]^{\text{cf}(\alpha)}, \\ A_\sigma &\in [\alpha]^{<\alpha_\sigma} \text{ for } \sigma \in \Gamma, \\ A_\sigma \cap A_{\sigma'} &= \emptyset \text{ for } \sigma, \sigma' \in \Gamma, \sigma \neq \sigma', \text{ and} \\ R(U^\xi) \cap R(U^{\xi'}) &= J_\sigma \text{ for } \xi, \xi' \in A_\sigma, \xi \neq \xi', \sigma \in \Gamma \\ &= J \text{ for } \xi \in A_\sigma, \xi' \in A_{\sigma'}, \sigma, \sigma' \in \Gamma, \sigma \neq \sigma'. \end{aligned}$$

We set

$$A = \bigcup_{\sigma \in \Gamma} A_\sigma \text{ and } \tilde{J} = \bigcup_{\sigma \in \Gamma} J_\sigma,$$

and we note that $|A| = \alpha$ and $|\tilde{J}| \leq \text{cf}(\alpha)$.

(a) There is $B \in [A]^\beta$ such that if $C \in [B]^\gamma$ then $\bigcap_{\xi \in C} \pi_J[U^\xi] \neq \emptyset$. Let $C \in [B]^\gamma$ and $p \in \bigcap_{\xi \in C} \pi_J[U^\xi]$, and set $F = \bigcup_{\xi \in C} R(U^\xi)$. Since

$$\begin{aligned} |\{\xi \in C : i \in R(U^\xi)\}| &\leq 1 \text{ for } i \in F \setminus \tilde{J} \text{ and} \\ |F \cup \tilde{J}| &\leq \gamma \cdot \kappa + \text{cf}(\alpha) = \gamma \cdot \text{cf}(\alpha), \end{aligned}$$

there is $x = \langle x_i : i \in I \rangle \in Y$ such that

$$\begin{aligned} x_i &= p_i \text{ if } i \in \tilde{J} \\ x_i &\in U_i^\xi \text{ if } i \in F \setminus \tilde{J}, i \in R(U^\xi) \\ x_i &\in X_i \text{ if } i \in I \setminus (F \cup \tilde{J}). \end{aligned}$$

It is clear that $x \in \bigcap_{\xi \in C} U^\xi$.

(b) There is $B \in [A]^\beta$ such that for $C \in [B]^\gamma$ there is compact $K_J \subset (X_J)_\kappa$ such that $\pi_J[U^\xi] \cap K_J \neq \emptyset$ whenever $\xi \in C$. Let $C \in [B]^\gamma$, let K_J be such a compact subspace of $(X_J)_\kappa$, and set $F = \bigcup_{\xi \in C} R(U^\xi)$. Since

$$|\{\xi \in C : i \in R(U^\xi)\}| \leq 1 \text{ for } i \in F \setminus \tilde{J},$$

there is $q \in X_{F \cup \tilde{J}}$ such that

$$q_i \in U_i^\xi \text{ if } i \in F \setminus \tilde{J}, i \in R(U^\xi).$$

We set $K_{F \cup J} = \{q\} \times K_J$. Then $K_{F \cup J}$ is by Lemma 3.2(b) a compact subspace of $(X_{F \cup J})_\kappa$, and since

$$|F \cup \tilde{J}| \leq \gamma \cdot \kappa + \text{cf}(\alpha) = \gamma \cdot \text{cf}(\alpha),$$

there is compact $K \subset Y$ such that $\pi_{F \cup J}[K] = K_{F \cup J}$. It is clear that $U^\xi \cap K \neq \emptyset$ whenever $\xi \in C$, as required.

(c) There is $p \in X_J$ such that

$$|\{\xi < \alpha : V_J \cap \pi_J[U^\xi] \neq \emptyset\}| \geq \beta$$

for every neighborhood V_J of p , and since $|\tilde{J}| \leq \text{cf}(\alpha)$ there is $x \in Y$ such that $x_J = p$. Let V be a basic neighborhood in $(X_I)_\kappa$ of x , and set $V_J = \pi_J[V]$,

$$C = \{\xi < \alpha : V_J \cap \pi_J[U^\xi] \neq \emptyset\} \text{ and } F = \bigcup_{\xi \in C} R(U^\xi).$$

If $\xi \in C$ and $V \cap U^\xi = \emptyset$, then from Lemma 3.1 we have $(R(V) \cap R(U^\xi)) \setminus \tilde{J} \neq \emptyset$. Since

$$|\{\xi \in C : i \in R(U^\xi) \setminus \tilde{J}\}| \leq 1 \text{ for } i \in F,$$

we have

$$|\{\xi \in C : V \cap U^\xi = \emptyset\}| \leq |R(V)| < \kappa \leq \beta,$$

and from $|C| \geq \beta$ we have

$$|\{\xi \in C : V \cap U^\xi \neq \emptyset\}| \geq \beta,$$

as required.

In the proof of part (b) of Theorem 3.13 we consider (implicitly) a set $\{X_i : i \in I\}$ of spaces, a set $\{\tilde{J}_\sigma : \sigma \in \Delta\}$ of pairwise disjoint, non-empty subsets of I , and a subspace $(\prod_{\sigma \in \Delta} K_\sigma)_\kappa$ of $(\prod_{i \in I} X_i)_\kappa$ (with $\tilde{I} = \cup_{\sigma \in \Delta} \tilde{J}_\sigma$) with K_σ a compact subspace of $(X_{\tilde{J}_\sigma})_\kappa$. Except in trivial cases, the product $(\prod_{\sigma \in \Delta} K_\sigma)_\kappa$ is not compact when $\kappa > \omega$. This explains the condition $\kappa = \omega$ in the statement of Theorem 3.13(b).

3.13 Theorem. Let α be singular with $\text{cf}(\alpha) > \omega$, let $\omega \leq \kappa \ll \alpha$, $\kappa \ll \text{cf}(\alpha)$, $\alpha \geq \beta \geq \omega$ and $\alpha \geq \beta \geq \gamma \geq 2$, and let $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ be a set of cardinal numbers such that

$$\begin{aligned} \alpha_\sigma &\text{ is regular for } \sigma < \text{cf}(\alpha), \\ \kappa &\ll \alpha_\sigma \text{ for } \sigma < \text{cf}(\alpha), \\ \text{cf}(\alpha) &< \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma &= \alpha. \end{aligned}$$

Let $\{\beta_\sigma : \sigma < \text{cf}(\alpha)\}$ be a set of cardinal numbers such that

$$\begin{aligned} \alpha_\sigma &\geq \beta_\sigma \geq 2 \text{ for } \sigma < \text{cf}(\alpha), \\ \beta_{\sigma'} &\leq \beta_\sigma \leq \beta \text{ for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \beta &= \sum_{\sigma < \text{cf}(\alpha)} \beta_\sigma, \end{aligned}$$

and let $\gamma_\sigma = \min\{\gamma, \beta_\sigma\}$ for $\sigma < \text{cf}(\alpha)$.

Let $\{X_i : i \in I\}$ be a set of non-empty spaces and $Y \subset (X_I)_\kappa$.

(a) If $(X_J)_\kappa$ has calibre (β, β, γ) and calibre $(\alpha_\sigma, \beta_\sigma, \gamma_\sigma)$ for every non-empty $J \subset I$ with $|J| < \kappa$, and if $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| \leq \gamma \cdot \kappa$, then Y has calibre (α, β, γ) ;

(b) if $\kappa = \omega$ and X_J has compact-calibre (β, β, γ) and compact-calibre $(\alpha_\sigma, \beta_\sigma, \gamma_\sigma)$ for every non-empty $J \subset I$ with $|J| < \omega$, and if for every compact $K_J \subset X_J$ with J non-empty, $J \subset I$ and $|J| \leq \gamma \cdot \omega$ there is compact $K \subset Y$ such that $\pi_J[K] = K_J$, then Y has compact calibre (α, β, γ) .

Proof. We assume in what follows, the statements being obvious otherwise, that $X_i \neq \emptyset$ for $i \in I$.

Let $\{U^\xi : \xi < \alpha\}$ be a set of non-empty basic subsets of $(X_J)_\kappa$. We show in (a) there is $B \in [\alpha]^\beta$ such that if $C \in [B]^\gamma$ then $(\bigcap_{\xi \in C} U^\xi) \cap Y \neq \emptyset$, and in (b) there is $B \in [\alpha]^\beta$ such that if $C \in [B]^\gamma$ there is compact $K \subset Y$ such that $U^\xi \cap K \neq \emptyset$ whenever $\xi \in C$.

From the extension to singular cardinals of the Erdős–Rado theorem on quasi-disjoint families (Theorem 1.10), there are Γ, J , $\{A_\sigma : \sigma \in \Gamma\}$ and $\{J_\sigma : \sigma \in \Gamma\}$ such that

$$\begin{aligned} \Gamma &\in [\text{cf}(\alpha)]^{\text{cf}(\alpha)}, \\ A_\sigma &\in [\alpha]^{a_\sigma} \text{ for } \sigma \in \Gamma, \\ A_\sigma \cap A_{\sigma'} &= \emptyset \text{ for } \sigma, \sigma' \in \Gamma, \sigma \neq \sigma' \text{ and} \\ R(U^\xi) \cap R(U^{\xi'}) &= J_\sigma \text{ for } \xi, \xi' \in A_\sigma, \xi \neq \xi' \\ &= J \text{ for } \xi \in A_\sigma, \xi' \in A_{\sigma'}, \sigma \neq \sigma'. \end{aligned}$$

We set $\tilde{J}_\sigma = J_\sigma \setminus J$ for $\sigma \in \Gamma$. If $\tilde{J}_\sigma = \emptyset$ we let B_σ be an arbitrary element of $[A_\sigma]^{\beta_\sigma}$, and if $\tilde{J}_\sigma \neq \emptyset$ then: in (a), using the fact that $(X_{J_\sigma})_\kappa$ has calibre $(\alpha_\sigma, \beta_\sigma, \gamma_\sigma)$, we choose $[B_\sigma] \in [A_\sigma]^{\beta_\sigma}$ such that if $C_\sigma \in [B_\sigma]^{\gamma_\sigma}$ then $\bigcap_{\xi \in C_\sigma} \pi_{J_\sigma}[U^\xi] \neq \emptyset$; and in (b), using the fact that X_{J_σ} has compact-calibre $(\alpha_\sigma, \beta_\sigma, \gamma_\sigma)$, we choose $B_\sigma \in [A_\sigma]^{\beta_\sigma}$ such that if $C_\sigma \in [B_\sigma]^{\gamma_\sigma}$ then there is compact $K_\sigma \subset X_{J_\sigma}$ such that $\pi_{J_\sigma}[U^\xi] \cap K_\sigma \neq \emptyset$ whenever $\xi \in C_\sigma$.

We set $\tilde{B} = \bigcup_{\sigma \in \Gamma} B_\sigma$. Since $\beta_{\sigma'} \leq \beta_\sigma$ for $\sigma' < \sigma < \text{cf}(\alpha)$ and $\beta = \sum_{\sigma < \text{cf}(\alpha)} \beta_\sigma$, we have $|\tilde{B}| = \beta$. If $J = \emptyset$ we let B be an arbitrary element of $|\tilde{B}|^\beta$, and if $J \neq \emptyset$ then: in (a), using the fact that $(X_J)_\kappa$ has calibre (β, β, γ) , we choose $B \in |\tilde{B}|^\beta$ such that if $C \in [B]^\gamma$ then $\bigcap_{\xi \in C} \pi_J[U^\xi] \neq \emptyset$; and in (b), using the fact that X_J has compact-calibre (β, β, γ) , we choose $B \in |\tilde{B}|^\beta$ such that if $C \in [B]^\gamma$ then there is compact $K \subset X_J$ such that $\pi_J[U^\xi] \cap K \neq \emptyset$ whenever $\xi \in C$.

To show that B is as required we let $C \in [B]^\gamma$ and we conclude separately the arguments for (a) and (b).

(a) If $\tilde{J}_\sigma \neq \emptyset$, then since

$$|C \cap B_\sigma| \leq \min\{\gamma, \beta_\sigma\} = \gamma_\sigma,$$

there is $p(\sigma) \in \{\pi_{J_\sigma}[U^\xi] : \xi \in C \cap B_\sigma\}$; and if $J \neq \emptyset$ there is $p \in \bigcap_{\xi \in C} \pi_J[U^\xi]$. We set $F = \bigcup_{\xi \in C} R(U^\xi)$; since $|F| \leq \gamma \cdot \kappa$ we have $\pi_F[Y] = X_F$, and since

$$|\{\xi \in C : i \in R(U^\xi)\}| \leq 1 \text{ for } i \in F \setminus \bigcup_{\sigma \in \Gamma} J_\sigma,$$

there is $x = \langle x_i : i \in I \rangle \in Y$ such that

$$\begin{aligned} x_i &= p_i && \text{if } i \in J \\ x_i &= p(\sigma)_i && \text{if } i \in \tilde{J}_\sigma, \sigma \in \Gamma \\ x_i &\in U_i^\xi && \text{if } i \in F \setminus \bigcup_{\sigma \in \Gamma} J_\sigma, i \in R(U^\xi), \xi \in C \\ x_i &\in X_i && \text{if } i \in I \setminus F. \end{aligned}$$

It is clear that $x \in (\cap_{\xi \in C} U^\xi) \cap Y$.

(b) If $\tilde{J}_\sigma \neq \emptyset$, then since

$$|C \cap B_\sigma| \leq \min \{\gamma, \beta_\sigma\} = \gamma_\sigma,$$

there is compact $K_\sigma \subset X_{J_\sigma}$ such that $\pi_{J_\sigma}[U^\xi] \cap K_\sigma \neq \emptyset$ whenever $\xi \in C \cap B_\sigma$; and if $J \neq \emptyset$ there is compact $K_J \subset X_J$ such that $\pi_J[U^\xi] \cap K_J \neq \emptyset$ whenever $\xi \in C$. We set $F = \bigcup_{\xi \in C} R(U^\xi)$, we note that

$$|\{\xi \in C : i \in R(U^\xi)\}| \leq 1 \text{ for } i \in F \setminus \bigcup_{\sigma \in \Gamma} J_\sigma,$$

and we choose

$$x_i \in U_i^\xi \text{ for } i \in F \setminus \bigcup_{\sigma \in \Gamma} J_\sigma, i \in R(U^\xi), \xi \in C.$$

Since $|F| \leq \gamma \cdot \omega$, there is compact $K \subset Y$ such that

$$\begin{aligned} \pi_J[K] &= K_J \text{ if } J \neq \emptyset \\ \pi_{\tilde{J}_\sigma}[K] &= K_\sigma \text{ if } \tilde{J}_\sigma \neq \emptyset \\ \pi_i[K] &= \{x_i\} \text{ if } i \in F \setminus \bigcup_{\sigma \in \Gamma} J_\sigma, i \in R(U^\xi). \end{aligned}$$

It is clear that $U^\xi \cap K \neq \emptyset$ whenever $\xi \in C$.

The proof is complete.

3.14 Corollary. Let α be singular with $\text{cf}(\alpha) > \omega$, let $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ be a set of cardinal numbers such that

$$\begin{aligned} \alpha_\sigma &\text{ is regular for } \sigma < \text{cf}(\alpha), \\ \text{cf}(\alpha) &< \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma &= \alpha, \end{aligned}$$

and let $\{X_i : i \in I\}$ be a set of spaces.

(a) If X_J has calibre α and calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$ for all non-empty $J \subset I$ with $|J| < \omega$, then X_I has calibre α ;

(b) if X_J has compact-calibre α and compact-calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$ for all non-empty $J \subset I$ with $|J| < \omega$, then X_I has compact-calibre α .

Proof. Theorem 3.13 reduces to this when $\kappa, \alpha, \beta, \gamma, \langle \alpha_\sigma, \beta_\sigma, \gamma_\sigma \rangle$ and Y are replaced by $\omega, \alpha, \alpha, \alpha, \langle \alpha_{\sigma+1}, \alpha_\sigma, \alpha_\sigma \rangle$ and X_I , respectively.

3.15 Corollary. Let α be singular with $\text{cf}(\alpha) > \omega$ and let $\{X_i : i \in I\}$ be a set of spaces.

(a) If each of the spaces X_i has calibre α and calibre β for all (infinite) regular $\beta < \alpha$, then X_I has calibre α ;

(b) if each of the spaces X_i has compact-calibre α and compact-calibre β for all (infinite) regular $\beta < \alpha$, then X_I has compact-calibre α .

Proof. (a) It follows from Theorem 2.2(a) that for all non-empty $J \subset I$ with $|J| < \omega$ the space X_J has calibre α and calibre β for all (infinite) regular $\beta < \alpha$. Part (a) then follows from Corollary 3.14(a).

The proof of (b) is similar.

3.16 Theorem. Let $\omega \leq \kappa \leq \alpha, \kappa \leq \text{cf}(\alpha)$, let $\{X_i : i \in I\}$ be a set of spaces and $Y \subset (X_I)_\kappa$. If $(X_J)_\kappa$ is pseudo- (α, α) -compact for every non-empty $J \subset I$ with $|J| < \kappa$, and if $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| < \kappa$, then Y is pseudo- (α, α) -compact.

Proof. We assume in what follows, the statement being obvious otherwise, that $X_i \neq \emptyset$ for $i \in I$.

If α is regular the theorem is the case $\beta = \alpha$ of Theorem 3.6. We assume that α is singular, using Theorem A.7 we choose a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ of regular cardinal numbers such that

$$\begin{aligned} \kappa \leq \alpha_\sigma < \alpha \text{ for } \sigma < \text{cf}(\alpha) \text{ and} \\ \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma = \alpha, \end{aligned}$$

we let $\{U^\xi : \xi < \alpha\}$ be a set of non-empty basic subsets of $(X_I)_\kappa$, and from the extension to singular cardinals of the Erdős–Rado theorem we choose $\Gamma, J, \{A_\sigma : \sigma \in \Gamma\}, A$ and $\{J_\sigma : \sigma \in \Gamma\}$ such that

$$\Gamma \in [\text{cf}(\alpha)]^{\text{cf}(\alpha)},$$

$$A_\sigma \in [\alpha]^{\alpha_\sigma} \text{ for } \sigma < \text{cf}(\alpha),$$

$$A_\sigma \cap A_{\sigma'} = \emptyset \text{ for } \sigma, \sigma' \in \Gamma, \sigma \neq \sigma',$$

$$A = \bigcup_{\sigma < \text{cf}(\alpha)} A_\sigma, \text{ and}$$

$$R(U^\xi) \cap R(U^{\xi'}) = J_\sigma \text{ for } \xi, \xi' \in A_\sigma, \xi \neq \xi'$$

$$= J \text{ for } \xi \in A_\sigma, \xi' \in A_{\sigma'}, \sigma, \sigma' \in \Gamma, \sigma \neq \sigma'.$$

If $J = \emptyset$ we let x be an arbitrary element of Y , and if $J \neq \emptyset$ we use the facts that $(X_J)_\kappa$ is pseudo- (α, α) -compact and $\pi_J[Y] = X_J$ to find $x \in Y$ such that

$$|\{\xi \in A : V_J \cap \pi_J[U^\xi] \neq \emptyset\}| = \alpha$$

for every neighborhood V_J in $(X_J)_\kappa$ of x_J .

Let V be a basic neighborhood in $(X_I)_\kappa$ of x and set $V_J = \pi_J[V]$ and $B = \{\xi \in A : V_J \cap \pi_J[U^\xi] \neq \emptyset\}$.

For $i \in I$ we set $C(i) = \{\xi \in B : i \in R(U^\xi)\}$. We note that if $i \in I$ and $\sigma \in \Gamma$ and $|C(i) \cap A_\sigma| \geq 2$ then $i \in J_\sigma$. Since $J_\sigma \cap J_{\sigma'} = J$ for $\sigma, \sigma' \in \Gamma$ with $\sigma \neq \sigma'$ it follows that

$$|\{\sigma \in \Gamma : |C(i) \cap A_\sigma| \geq 2\}| \leq 1 \text{ for } i \in I \setminus J.$$

Since $C(i) = \bigcup_{\sigma \in \Gamma} (C(i) \cap A_\sigma)$ and $|A_\sigma| = \alpha_\sigma < \alpha$ and $|\Gamma| = \text{cf}(\alpha) < \alpha$, we have $|C(i)| < \alpha$ for $i \in I \setminus J$.

We set $C = \bigcup \{C(i) : i \in R(V) \setminus J\}$. Since $|R(V)| < \kappa < \text{cf}(\alpha)$ we have $|C| < \alpha$, and from $|B| = \alpha$ it follows that $|B \setminus C| = \alpha$.

If $\xi \in B$ and $V \cap U^\xi = \emptyset$, then since $V_J \cap \pi_J[U^\xi] \neq \emptyset$ there is $i \in I$ such that

$$i \in (R(V) \cap R(U^\xi)) \setminus J,$$

and hence $\xi \in C(i) \subset C$. We have

$$\{\xi \in B : V \cap U^\xi \neq \emptyset\} \supset B \setminus C$$

and $|B \setminus C| = \alpha$, as required.

Chain conditions in powers

3.17 Lemma. Let $\omega \leq \text{cf}(\alpha) < \text{cf}(\beta) \leq \beta < \alpha$ and let X be a space.

(a) If X has calibre α then there is γ such that $\beta < \gamma < \alpha$ and X has calibre (γ, β) ; and

(b) if X has compact-calibre α then there is γ such that $\beta < \gamma < \alpha$ and X has compact-calibre (γ, β) .

Proof. We prove (b). If the result fails there is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ of cardinal numbers such that

$$\beta = \alpha_0 < \alpha_{\sigma'} < \alpha_\sigma < \alpha \quad \text{for } 0 < \sigma' < \sigma < \text{cf}(\alpha),$$

$$\alpha = \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma, \text{ and}$$

$$X \text{ does not have compact-calibre } (\alpha_\sigma, \beta) \text{ for } \sigma < \text{cf}(\alpha).$$

There is $\{U_\xi : \alpha_\sigma \leq \xi < \alpha_{\sigma+1}\} \subset \mathcal{T}^*(X)$ such that $|\{\xi : \alpha_\sigma \leq \xi < \alpha_{\sigma+1}, U_\xi \cap K \neq \emptyset\}| < \beta$ for all compact $K \subset X$; since $\{U_\xi : \beta \leq \xi < \alpha\} \subset \mathcal{T}^*(X)$ and X has compact-calibre (α, β) , there are $B \in [\alpha \setminus \beta]^\beta$ and compact $K \subset X$ such that $U_\xi \cap K \neq \emptyset$ whenever $\xi \in B$. We set

$$B(\sigma) = B \cap [\alpha_\sigma, \alpha_{\sigma+1}) \text{ for } \alpha < \text{cf}(\alpha).$$

Since $B = \bigcup_{\sigma < \text{cf}(\alpha)} B(\sigma)$ and $\text{cf}(\alpha) < \text{cf}(\beta)$, there is $\bar{\sigma} < \text{cf}(\alpha)$ such that $|B(\bar{\sigma})| = \beta$. This contradiction completes the proof.

We omit the (similar, simpler) proof of (a).

A non-empty finite space has calibre α for all $\alpha \geq \omega$; but from Corollary 2.10 and Theorem 2.3(a) it follows that if $\text{cf}(\alpha) = \omega$ then for $\gamma \geq \omega$ the (infinite) space $\{0, 1\}^\gamma$ does not have calibre α . We show next, in contrast, that for $\text{cf}(\alpha) > \omega$ every power of a space with calibre α has calibre α .

3.18 Theorem. Let $\text{cf}(\alpha) > \omega$ and let X be a space.

(a) If X has calibre α then X^I has calibre α for every non-empty set I ; and

(b) if X has compact-calibre α then X^I has compact-calibre α for every non-empty set I .

Proof. For α regular this is a special case of Corollary 3.7. We assume in what follows that α is singular.

There is a set $\{\beta_\sigma : \sigma < \text{cf}(\alpha)\}$ of cardinal numbers such that

$$\begin{aligned} \beta_{\sigma'} < \beta_\sigma < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha) \text{ and} \\ \sum_{\sigma < \text{cf}(\alpha)} \beta_\sigma &= \alpha. \end{aligned}$$

We define sets $\{\alpha_\sigma(n) : \sigma < \text{cf}(\alpha), n < \omega\}$ and $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ of cardinal numbers such that

- (i) $\alpha_0 = \alpha_0(n) = (\text{cf}(\alpha))^+$ for $n < \omega$;
- (ii) $\alpha_\sigma = \alpha_\sigma(n) = (\max\{\beta_\sigma, \sup_{\sigma' < \sigma} \alpha_{\sigma'}\})^+$ for non-zero limit ordinals $\sigma < \text{cf}(\alpha)$;
- (iii) $\alpha_{\sigma+1}(n) > \alpha_\sigma$ for $\sigma < \text{cf}(\alpha), n < \omega$;
- (iv)(a) in (a): X^n has calibre $(\alpha_{\sigma+1}(n), \alpha_\sigma)$ for $\sigma < \text{cf}(\alpha), n < \omega$;
- (iv)(b) in (b): X^n has compact-calibre $(\alpha_{\sigma+1}(n), \alpha_\sigma)$ for $\sigma < \text{cf}(\alpha), n < \omega$; and

$$(v) \alpha_{\sigma+1} = (\max\{\beta_{\sigma+1}, \sup_{n < \omega} \alpha_{\sigma+1}(n)\})^+ \text{ for } \sigma < \text{cf}(\alpha).$$

We proceed by transfinite recursion. We define $\alpha_0, \alpha_\sigma(n)$ for $n < \omega$

by (i), and for non-zero limit ordinals $\sigma < \text{cf}(\alpha)$ we define $\alpha_\sigma, \alpha_\sigma(n)$ for $n < \omega$ by (ii). For $\sigma < \text{cf}(\alpha)$ we note that α_σ is a regular cardinal, we note for $n < \omega$ from Theorem 2.2 that X^n has calibre α in (a) and X^n has compact-calibre α in (b), and we use Lemma 3.17 to define $\alpha_{\sigma+1}(n)$ for $n < \omega$ so that (iii), and (iv)(a) or (iv)(b), are satisfied; and we define $\alpha_{\sigma+1}$ by (v).

The definition of the families is complete. We note that:

α_σ is regular for $\sigma < \text{cf}(\alpha)$;

$\text{cf}(\alpha) < \alpha_{\sigma'} < \alpha_\sigma < \alpha$ for $\sigma' < \sigma < \text{cf}(\alpha)$;

$\sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma = \alpha$; and

X_J has calibre α and calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$ for all $J \in \mathcal{P}_\omega^*(I)$ in (a), and

X_J has compact-calibre α and compact-calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$ for all $J \in \mathcal{P}_\omega^*(I)$ in (b).

The required conclusions now follow from Corollary 3.14.

Remark. We have seen in Corollary 3.7(a) that for $\alpha > \omega$ with α regular the product of a set of spaces with calibre α has calibre α . For singular α the analogous statement is false (cf. Theorem 4.4); thus in Theorem 3.18(a) the restriction to powers of a fixed space cannot be waived, and Theorem 3.18(a) is, in this sense, the best possible. For compact spaces, however, a strong positive result is available (Corollary 5.8): Assuming the generalized continuum hypothesis, the product of a set of compact spaces with calibre α (with $\text{cf}(\alpha) > \omega$) has calibre α .

In Theorem 3.19 we generalize the results of Theorem 3.18.

3.19 Theorem. Let $\omega \leq \kappa \ll \alpha, \kappa \ll \text{cf}(\alpha)$, $\alpha \geq \beta \geq \gamma \geq 2$, and let X be a space.

(a) If $(X^J)_\kappa$ has calibre (α, β, γ) for all J with $|J| < \kappa$, then $(X^I)_\kappa$ has calibre (α, β, γ) for all I ;

(b) if $(\kappa = \omega \text{ and } X^n \text{ has compact-calibre } (\alpha, \beta, \gamma) \text{ for all } n < \omega)$, then X^I has compact-calibre (α, β, γ) for all I ;

(c) if $(\kappa = \omega \text{ and } X^n \text{ is pseudo-}(\alpha, \beta)\text{-compact for all } n < \omega)$, then X^I is pseudo- (α, β) -compact for all I .

Proof. The statements are clear if $X = \emptyset$; we assume in what follows that $X \neq \emptyset$.

Let $\{U^\xi: \xi < \alpha\}$ be a set of non-empty, basic open subsets of $(X^I)_\kappa$.

For notational convenience we let $R(\xi)$ denote $R(U^\xi)$ for $\xi < \alpha$. Since $|R(\xi)| < \kappa$ for $\xi < \alpha$ and $\kappa \ll \text{cf}(\alpha)$ we may suppose without loss of generality that there is $\lambda < \kappa$ such that $|R(\xi)| = \lambda$ for all $\xi < \alpha$; for $\xi < \alpha$ we let φ_ξ be a one-to-one function from $R(\xi)$ onto λ and, using Corollary 1.11, we choose $A \in [\alpha]^\alpha$ such that if $\xi, \xi' \in A$ and $i \in R(\xi) \cap R(\xi')$ then $\varphi_\xi(i) = \varphi_{\xi'}(i)$.

For $\xi \in A$ we define a function $h_\xi: X^{R(\xi)} \rightarrow X^\lambda$ by the rule

$$(h_\xi(x))_\zeta = x_i \quad \text{for } \zeta = \varphi_\xi(i) < \lambda.$$

It is clear that h_ξ is a homeomorphism of $(X^{R(\xi)})_\kappa$ onto $(X^\lambda)_\kappa$.

For $\xi \in A$ we set $V^\xi = \pi_{R(\xi)}[U^\xi]$. We complete the proofs of (a) and (b) separately.

(a) Since $h_\xi[V^\xi]$ is open in $(X^\lambda)_\kappa$ and $(X^\lambda)_\kappa$ has calibre (α, β, γ) , there is $B \in [A]^B$ such that if $C \in [B]^\gamma$ then $\bigcap_{\xi \in C} h_\xi[V^\xi] \neq \emptyset$. We claim that if $C \in [B]^\gamma$ then $\bigcap_{\xi \in C} U^\xi \neq \emptyset$. Let $C \in [B]^\gamma$ and $p \in \bigcap_{\xi \in C} h_\xi[V^\xi]$ and define $x \in X^I$ by the rule

$$\begin{aligned} x_{R(\xi)} &= h_\xi^{-1}(\{p\}) \quad \text{for } \xi \in C, \\ x_i &\in X_I \quad \text{if } i \in I \setminus \bigcup_{\xi \in C} R(\xi). \end{aligned}$$

(If $i \in R(\xi) \cap R(\xi')$ with $\xi, \xi' \in C$, there is $\zeta < \lambda$ such that

$$\varphi_\xi(i) = \varphi_{\xi'}(i) = \zeta;$$

hence $x_i = p_\zeta$. It follows that x is a well-defined element of X^I .)

For $\xi \in C$ we have $h_\xi(x_{R(\xi)}) = p \in h_\xi[V^\xi]$ and hence $x_{R(\xi)} \in V^\xi$. Since $U_i^\xi = X$ for $i \in I \setminus R(\xi)$ we have $x \in U^\xi$ for $\xi \in C$, as required.

(b) Since $h_\xi[V^\xi]$ is open in X^λ and X^λ has compact-calibre (α, β, γ) , there is $B \in [A]^\beta$ such that if $C \in [B]^\gamma$ there is compact $\tilde{K} \subset X^\lambda$ such that $h_\xi[V^\xi] \cap \tilde{K} \neq \emptyset$ whenever $\xi \in C$. We claim that if $C \in [B]^\gamma$ there is compact $K \subset X^I$ such that $U^\xi \cap K \neq \emptyset$ whenever $\xi \in C$. Let $C \in [B]^\gamma$ and let $\tilde{K}$ be a compact subspace of X^λ such that $h_\xi[V^\xi] \cap \tilde{K} \neq \emptyset$ whenever $\xi \in C$; we suppose without loss of generality that $\tilde{K} = \prod_{k < \lambda} \pi_k[\tilde{K}]$.

We choose $x \in X$ and we define $K = \prod_{i \in I} K_i \subset X^I$ by the rule

$$\prod_{i \in R(U^\xi)} K_i = h_\xi^{-1}(\tilde{K}) \quad \text{for } \xi \in C,$$

$$K_i = \{x\} \quad \text{if } i \in I \setminus \bigcup_{\xi \in C} R(\xi).$$

(If $i \in R(\xi) \cap R(\xi')$ with $\xi, \xi' \in C$, there is $k < \lambda$ such that

$$\varphi_{\xi}(i) = \varphi_{\xi'}(i) = k;$$

hence $K_i = \tilde{K}_k$. It follows that K is a well-defined (compact) subspace of X^I .)

For $\xi \in C$ we have

$$h_{\xi}[V^{\xi}] \cap h_{\xi}[\pi_{R(\xi)}[K]] = h_{\xi}[V_{\xi}] \cap \tilde{K} \neq \emptyset$$

and hence $V^{\xi} \cap \pi_{R(\xi)}[K] \neq \emptyset$. Since $U_i^{\xi} = X$ for $i \in I \setminus R(\xi)$, we have $U^{\xi} \cap K \neq \emptyset$ for $\xi \in C$, as required.

(c) Since $h_{\xi}[V^{\xi}]$ is open in X^{λ} and X^{λ} is pseudo- (α, β) -compact, there is $p \in X^{\lambda}$ such that

$$|\{\xi \in A : V \cap h_{\xi}[V^{\xi}] \neq \emptyset\}| \geq \beta$$

for every neighborhood V of p . We define $x \in X^I$ by the rule

$$\begin{aligned} X_{R(\xi)} &= h_{\xi}^{-1}(\{p\}) \quad \text{for } \xi \in A, \\ x_i &\in X_i \quad \text{if } i \in I \setminus \bigcup_{\xi \in A} R(\xi) \end{aligned}$$

and we show

$$|\{\xi \in A : W \cap U^{\xi} \neq \emptyset\}| \geq \beta$$

for every neighborhood W of x . (That x is a well-defined element of X^I follows as in the proof above of part (a).)

Let W be a basic neighborhood of x and let F be a finite subset of I such that

$$R(W) \cap \left(\bigcup_{\xi \in A} R(\xi) \right) \subset \bigcup_{\eta \in F} R(\eta).$$

Since $\pi_{R(\eta)}[W]$ is a neighborhood of $x_{R(\eta)}$ and h_{η} is a homeomorphism of $X^{R(\eta)}$ onto X^{λ} , the set $h_{\eta}[\pi_{R(\eta)}[W]]$ is a neighborhood of $h_{\eta}(x_{R(\eta)}) = p$. We set

$$V = \bigcap_{\eta \in F} h_{\eta}[\pi_{R(\eta)}[W]].$$

Since V is a neighborhood of p we have

$$|\{\xi \in A : V \cap h_{\xi}[V^{\xi}] \neq \emptyset\}| \geq \beta.$$

To complete the proof it is enough to show that if $V \cap h_{\xi}[V^{\xi}] \neq \emptyset$ with $\xi \in A$ then $W \cap U^{\xi} \neq \emptyset$, and for this (since $W_i = X$ for $i \in I \setminus \bigcup_{\eta \in F} R(\eta)$) it is enough to show that

$$\pi_{R(\eta)}[W] \cap \pi_{R(\eta)}[U^{\xi}] \neq \emptyset \quad \text{for } \eta \in F.$$

We have

$$\begin{aligned}
 h_\xi \circ \pi_{R(\xi)}[U^\xi] &= \prod_{\xi < \lambda} U_{\varphi_\xi^{-1}(\{\xi\})}^\xi \\
 &= \prod_{\xi \in \varphi_\xi[R(\xi) \cap R(\eta)]} U_{\varphi_\xi^{-1}(\{\xi\})}^\xi \times \prod_{\xi \in \varphi_\xi[R(\xi) \setminus R(\eta)]} U_{\varphi_\xi^{-1}(\{\xi\})}^\xi \\
 &\subset \prod_{\xi \in \varphi_\xi[R(\xi) \cap R(\eta)]} U_{\varphi_\xi^{-1}(\{\xi\})}^\xi \times \prod_{\xi \in \varphi_\xi[R(\xi) \setminus R(\eta)]} X \\
 &= \prod_{\xi \in \varphi_\eta[R(\xi) \cap R(\eta)]} U_{\varphi_\eta^{-1}(\{\xi\})}^\xi \times \prod_{\xi \in \varphi_\eta[R(\xi) \setminus R(\eta)]} X \\
 &= \prod_{\xi < \lambda} U_{\varphi_\eta^{-1}(\{\xi\})}^\xi = h_\eta \circ \pi_{R(\eta)}[U^\xi],
 \end{aligned}$$

and from the relations

$$\begin{aligned}
 \emptyset &= V \cap h_\xi[V^\xi] = V \cap h_\xi[\pi_{R(\xi)}[U^\xi]] \\
 &\subset V \cap h_\eta[\pi_{R(\eta)}[U^\xi]] \subset h_\eta[\pi_{R(\eta)}[W]] \cap h_\eta[\pi_{R(\eta)}[U^\xi]] \\
 &= h_\eta[\pi_{R(\eta)}[W] \cap \pi_{R(\eta)}[U^\xi]]
 \end{aligned}$$

we have $\pi_{R(\eta)}[W] \cap \pi_{R(\eta)}[U^\xi] \neq \emptyset$, as required.

3.20 Remark. We note that although Theorems 3.18 and 3.19 concern powers (of the form X^I or $(X^I)_\kappa$), similar results are available for products X_I or $(X_I)_\kappa$ which are sufficiently closely related to powers. We illustrate with a simple modification of Theorem 3.19(a).

Theorem. Let $\omega \leq \kappa \ll \alpha$, $\alpha \geq \beta \geq \gamma \geq 2$ and let $\{X_i : i \in I\}$ be a set of spaces. If there are a space Y and for $i \in I$ a continuous function from Y onto a dense subspace of X_i , and if $(Y^J)_\kappa$ has calibre (α, β, γ) for all J with $|J| < \kappa$, then $(X_I)_\kappa$ has calibre (α, β, γ) .

Proof. The space $(Y^I)_\kappa$ has calibre (α, β, γ) by Theorem 3.19(a), and there is a continuous function from $(Y^I)_\kappa$ onto a dense subspace of $(X_I)_\kappa$. That $(X_I)_\kappa$ has calibre (α, β, γ) follows from Theorem 2.5(a)(i) and (ii).

3.21 Remark. The methods of Theorem 3.19 do not apply when $\text{cf}(\alpha) = \omega$, and it is natural to ask the following three questions.

Let α be singular with $\text{cf}(\alpha) = \omega$ and $\alpha > \beta$ and let P be one of the properties: calibre (α, β) ; compact-calibre (α, β) ; pseudo- (α, β) -compact. Is there a completely regular, Hausdorff space X such that

X^n has P for all $n < \omega$, and

X^ω does not have P ?

(The corresponding question is also unanswered for compact-calibre (α, α) with α singular, cf $(\alpha) = \omega$.)

3.22 Theorem. Let κ and λ be cardinal numbers with $\kappa > \lambda \geq \omega$ and let $\{X_i : i \in I\}$ be a set of regular, Hausdorff spaces with $|I| \geq \lambda$ and with $|X_i| > 1$ for $i \in I$. Then $(X_I)_\kappa$ is not pseudo- $(2^\lambda, 2)$ -compact.

Proof. Let $\{i(\eta) : \eta < \lambda\} \subset I$ with $i(\eta) \neq i(\eta')$ for $\eta' < \eta < \lambda$, for $\eta < \lambda$ let $U_\eta, V_\eta, \tilde{U}_\eta, \tilde{V}_\eta$ be non-empty open subsets of $X_{i(\eta)}$ such that

$$\text{cl } U_\eta \subset \tilde{U}_\eta, \text{cl } V_\eta \subset \tilde{V}_\eta, \tilde{U}_\eta \cap \tilde{V}_\eta = \emptyset,$$

and set

$$W(A) = \left(\bigcap_{\eta \in A} \pi_{i(\eta)}^{-1}(U_\eta) \right) \cap \left(\bigcap_{\eta \in \lambda \setminus A} \pi_{i(\eta)}^{-1}(V_\eta) \right) \quad \text{for } A \subset \lambda.$$

Since $\kappa > \lambda$ the sets $W(A)$ are open in $(X_I)_\kappa$.

We show that for $x \in (X_I)_\kappa$ there is a neighborhood W of x such that

$$|\{A \subset \lambda : W \cap W(A) \neq \emptyset\}| \leq 1.$$

Case 1. $x_{i(\eta)} \in \text{cl } U_\eta \cup \text{cl } V_\eta$ for $\eta < \lambda$. We set

$$B = \{\eta \in \lambda : x_{i(\eta)} \in \text{cl } U_\eta\}, \text{ and}$$

$$W = \left(\bigcap_{\eta \in B} \pi_{i(\eta)}^{-1}(\tilde{U}_\eta) \right) \cap \left(\bigcap_{\eta \in \lambda \setminus B} \pi_{i(\eta)}^{-1}(\tilde{V}_\eta) \right).$$

Then for $A \subset \lambda$ we have $W \cap W(A) \neq \emptyset$ if and only if $A = B$.

Case 2. There is $\bar{\eta} < \lambda$ such that $x_{i(\bar{\eta})} \notin \text{cl } U_{\bar{\eta}} \cup \text{cl } V_{\bar{\eta}}$. We set

$$W = \pi_{i(\bar{\eta})}^{-1}(X_{i(\bar{\eta})} \setminus (\text{cl } U_{\bar{\eta}} \cup \text{cl } V_{\bar{\eta}})).$$

Then $W \cap W(A) = \emptyset$ for all $A \subset \lambda$.

The proof is complete.

It follows from Theorem 2.1(d) and (e) that a space $(X_I)_\kappa$ as in Theorem 3.22 does not have calibre $(2^\lambda, 2)$. For this latter conclusion it is not necessary to assume that the Hausdorff spaces X_i are regular spaces, since the procedure of the proof of Theorem 3.22 furnishes a cellular family in $(X_I)_\kappa$ faithfully indexed by $\{A : A \subset \lambda\}$.

3.23 Corollary. Let α, κ and λ be infinite cardinals with $\kappa > \lambda$

Chain Conditions in Products													
	calibre (α, β, γ)				compact-calibre (α, β, γ)				pseudo- (α, β) -compact				
	α regular $\alpha = \omega$ $\alpha > \omega$		α singular $\text{cf}(\alpha) = \omega$ $\text{cf}(\alpha) > \omega$		α regular $\alpha = \omega$ $\alpha > \omega$		α singular $\text{cf}(\alpha) = \omega$ $\text{cf}(\alpha) > \omega$		α regular $\alpha = \omega$ $\alpha > \omega$		α singular $\text{cf}(\alpha) = \omega$ $\text{cf}(\alpha) > \omega$		
$\alpha = \beta$	N	Y	N	N	Y	Y	N	N	N	Y	N	Y	
	2.10	3.6	4.4	4.4	3.9	3.6	8.3	8.3	9.9	3.6, 3.16	9.9	3.16	
Products	$\alpha > \beta$	N	Y	N	N	Y	Y	$\beta \geq \omega$ N 3.11 $\beta \geq \omega$ N 3.11 $\beta < \omega$ $\beta < \omega$ Y 2.1 (g) Y 2.1(g)		N	Y	N	N
		2.10	3.6	3.11	3.11	2.1(g)	3.6			9.9 + 2.1(i)	3.6	3.11	3.11
$\alpha = \beta$	N	Y	N	Y	Y	Y		Y	N	Y	N	Y	
	2.10	3.6	2.12	3.18, 3.19	3.9	3.6	?	3.18, 3.19	9.10	3.6, 3.16	9.10	3.16, 3.19	
Powers	$\alpha > \beta$	N	Y	?	Y	Y	Y	$\beta \geq \omega$? $\beta < \omega$ Y 2.1 (g) Y		N	Y	?	Y
		2.10	3.6		3.19	2.1(g)	3.6			9.10 + 2.1 (i)	3.6		3.19

and let $\{X_i : i \in I\}$ be a set of regular, Hausdorff spaces with $|I| \geq \lambda$ and with $|X_i| > 1$ for $i \in I$.

(a) If $2 \leq \beta \leq \alpha \leq 2^\lambda$ then $(X_I)_\kappa$ is not pseudo- (α, β) -compact.

(b) If $\text{cf}(\alpha) \leq 2^\lambda$ then $(X_I)_\kappa$ is not pseudo- (α, α) -compact.

Proof. (a) follows from Theorems 3.22 and 2.1(c).

(b) follows from (a) (with α and β replaced by $\text{cf}(\alpha)$) and Theorem 2.3(c).

We note that from Theorem 2.1(d) and (e) it follows that a space $(X_I)_\kappa$ as in Corollary 3.23(b) does not have compact-calibre α or calibre α .

We note further that the assumptions $\alpha \leq 2^\lambda$ and $\text{cf}(\alpha) \leq 2^\lambda$ of Corollary 3.23 are incompatible with the conditions $\kappa \ll \alpha$ and $\kappa \ll \text{cf}(\alpha)$, respectively, which have occurred frequently as hypotheses in this chapter.

3.24 Summary. Some of the results of this chapter and of later chapters are summarized in the chart opposite. For each of the indicated chain conditions P (with $\alpha \geq \beta \geq \gamma \geq 2$) the question asked is this: If $\{X_i : i \in I\}$ is a set of spaces such that $(X_J)_\kappa$ has P for all non-empty $J \subset I$ with $|J| < \kappa$, must $(X_I)_\kappa$ have P ?

Some of the results cited and some of the examples given relate only to the case $\kappa = \omega$. It is understood in each case that $\alpha \geq \kappa$, and frequently for $\kappa > \omega$ the theorems cited require in addition one or both of the hypotheses $\kappa \ll \alpha, \kappa \ll \text{cf}(\alpha)$. Numbers refer to the relevant passage in this text. Y indicates that the answer to the question is Yes; N indicates No.

The numbers $S_\lambda((X_I)_\kappa)$

3.25 Theorem. Let $\lambda \geq 1, \kappa \geq \omega$, let $\{X_i : i \in I\}$ be a set of non-empty spaces and

$$\alpha = \sup \{S_\lambda((X_J)_\kappa) : \emptyset \neq J \subset I, |J| < \kappa\},$$

and let β be a regular cardinal such that $\beta \geq \alpha$ and $\kappa \ll \beta$. Then

(a) $\alpha \leq S_\lambda((X_I)_\kappa) \leq \beta$; and

(b) if the spaces X_i are Hausdorff spaces and $\alpha < \omega$, then $S_\lambda((X_I)_\kappa) = \alpha$.

Proof. (a) For $J \in \mathcal{P}_\kappa^*(I)$ the space $(X_J)_\kappa$ is the continuous image of $(X_I)_\kappa$, and from Theorem 2.5(a)(i) we have $S_\lambda((X_J)_\kappa) \leq S_\lambda((X_I)_\kappa)$. It follows that $\alpha \leq S_\lambda((X_I)_\kappa)$.

That $S_\lambda((X_I)_\kappa) \leq \beta$ follows from Theorem 3.6(a) (with α, β, λ and Y replaced by β, λ, λ and $(X_I)_\kappa$, respectively).

(b) If $\lambda = 1$ then from Lemma 2.8(c) we have $S_\lambda((X_J)_\kappa) = 1$ for all $J \in \mathcal{P}_\kappa^*(I)$ and hence

$$\alpha = 1 = S_\lambda((X_I)_\kappa).$$

If $\lambda > 1$ then from Corollary 2.10 we have $|(X_J)_\kappa| < \omega$ for all $J \in \mathcal{P}_\kappa^*(I)$. If for all $n < \omega$ there is $J(n) \in \mathcal{P}_\kappa^*(I)$ such that $|(X_{J(n)})_\kappa| \geq n$, then $\alpha \geq S_\lambda((X_{J(n)})_\kappa) \geq S_2((X_{J(n)})_\kappa) \geq S_2(n) = n + 1$ and hence $\alpha \geq \omega$, a contradiction. It follows that there are $n < \omega$ and $\bar{J} \in \mathcal{P}_\kappa^*(I)$ such that $|\bar{J}| = n$ and $|X_i| = 1$ for $i \in I \setminus \bar{J}$. Then $(X_I)_\kappa$ is homeomorphic to $(X_{\bar{J}})_\kappa$ and

$$S_\lambda((X_I)_\kappa) = S_\lambda((X_{\bar{J}})_\kappa) = \alpha,$$

as required.

3.26 Remarks. We continue the notation of Theorem 3.25.

(a) If α is regular and $\kappa \ll \alpha$ then $S_\lambda((X_I)_\kappa) = \alpha$.

(b) If $\alpha \geq \omega$ and $\gamma = ((\alpha^\kappa)^\kappa)^+$ then γ is a regular cardinal such that $\gamma > \alpha$ and $\kappa \ll \gamma$; hence $S_\lambda((X_I)_\kappa) \leq \gamma$. (That γ is regular is clear. For $\delta < \gamma, \mu < \kappa$ we have

$$\delta^\mu \leq ((\alpha^\kappa)^\kappa)^\mu \leq ((\alpha^\kappa)^\kappa)^\kappa = (\alpha^\kappa)^\kappa < \gamma$$

from Theorem A.3(c) of the Appendix; hence $\kappa \ll \omega$.)

(c) It follows from Theorem 3.25 that if $\lambda = 2$ or λ is (infinite and) regular, and if X_I is infinite and the spaces X_i are Hausdorff spaces, then $S_\lambda((X_I)_\kappa)$ is regular. But even when the spaces X_i are completely regular, Hausdorff spaces the relation $\kappa \ll S_\lambda((X_I)_\kappa)$ can fail. For a trivial instance of this phenomenon let γ be an uncountable regular cardinal such that $\gamma \geq \lambda$ and such that $\kappa \ll \gamma$ fails, using Theorem 3.30 below let X_0 be a space such that $S_\lambda(X_0) = \lambda$, let I be a set such that $0 \in I$ and let $|X_i| = 1$ for $i \in I, i \neq 0$; then

$$S_\lambda((X_I)_\kappa) = S_\lambda(X_0) = \lambda.$$

(d) We give a natural condition sufficient to ensure that $\kappa \ll \alpha$. We begin with a simple lemma.

Lemma. Let $\kappa \geq \omega$, let I be a set, $\{J(\eta) : \eta < \kappa\} \subset \mathcal{P}_\kappa(I)$ and $\mu < \kappa$. Then there is $A \in [\kappa]^\mu$ such that $|\bigcup_{\eta \in A} J(\eta)| < \kappa$.

Proof. If κ is regular then $|\bigcup_{\eta \in A} J(\eta)| < \kappa$ for all $A \in [\kappa]^\mu$. We assume that κ is singular. There is a set $\{\kappa_\sigma : \sigma < \text{cf}(\kappa)\}$ of cardinal numbers such that

$$\kappa_{\sigma'} < \kappa_\sigma < \kappa \quad \text{for } \sigma' < \sigma < \text{cf}(\kappa), \text{ and} \\ \sum_{\sigma < \text{cf}(\kappa)} \kappa_\sigma = \kappa.$$

We set $A(\sigma) = \{\eta < \kappa : |J(\eta)| \leq \kappa_\sigma\}$ for $\sigma < \text{cf}(\kappa)$ and we consider two cases.

Case 1. There is $\sigma < \text{cf}(\kappa)$ such that $|A(\sigma)| \geq \mu$. We choose $A \in [A(\sigma)]^\mu$ and we note that $|\bigcup_{\eta \in A} J(\eta)| \leq \mu \cdot \kappa_\sigma < \kappa$, as required.

Case 2. $|A(\sigma)| < \mu$ for all $\sigma < \text{cf}(\kappa)$. Then since $\kappa = \bigcup_{\sigma < \text{cf}(\kappa)} A(\sigma)$ we have

$$\kappa \leq \sum_{\sigma < \text{cf}(\kappa)} |A(\sigma)| \leq (\text{cf}(\kappa)) \cdot \mu < \kappa,$$

a contradiction.

Theorem. Let $\kappa \geq \omega$ and let λ be a cardinal such that either

- (i) $\lambda = 2$, or
- (ii) $\kappa \leq \text{cf}(\lambda)$, and if $\bar{\lambda} < \lambda$ and $\mu < \kappa$ then $\bar{\lambda}^\mu < \lambda$.

Let $\{X_i : i \in I\}$ be a set of non-empty spaces and

$$\alpha = \sup \{S_\lambda((X_J)_\kappa) : J \in \mathcal{P}_\kappa^*(I)\},$$

and suppose for all $\gamma < \alpha$ there is $\{J(\eta) : \eta < \kappa\} \subset \mathcal{P}_\kappa^*(I)$ such that

$$J(\eta') \cap J(\eta) = \emptyset \quad \text{for } \eta' < \eta < \kappa \text{ and}$$

$$S_\lambda((X_{J(\eta)})_\kappa) > \gamma.$$

If $\gamma < \alpha$ and $\mu < \kappa$ then $\gamma^\mu < \alpha$.

Proof. Let $\gamma < \alpha$ and $\mu < \kappa$ and let $\{J(\eta) : \eta < \kappa\} \subset \mathcal{P}_\kappa^*(I)$ satisfy

$$J(\eta') \cap J(\eta) = \emptyset \quad \text{for } \eta' < \eta < \kappa \text{ and}$$

$$S_\lambda((X_{J(\eta)})_\kappa) > \gamma \quad \text{for } \eta < \kappa.$$

For $\eta < \kappa$ there is $\{U^\xi(\eta) : \xi < \gamma\} \subset \mathcal{T}^*((X_{J(\eta)})_\kappa)$ such that $\bigcap_{\xi \in C} U^\xi(\eta) = \emptyset$ for all $C \in [\gamma]^\lambda$; from the lemma there is $A \in [\kappa]^\mu$ such that $|\bigcup_{\eta \in A} J(\eta)| < \kappa$.

We set $\bar{J} = \bigcup_{\eta \in A} J(\eta)$. For $f \in \gamma^A$ we set $U(f) = \prod_{\eta \in A} U^{f(\eta)}(\eta)$ and we note that $U(f) \in \mathcal{T}^*((X_{\bar{J}})_\kappa)$.

For $C \in [\gamma^A]^\lambda$ and $\eta \in A$ we set

$$C(\eta) = \{f(\eta) : f \in C\}.$$

We claim that if $C \in [\gamma^A]^\lambda$ there is $\bar{\eta} \in A$ such that $|C(\bar{\eta})| = \lambda$. We consider two cases.

Case 1. $\lambda = 2$. Let $C = \{f_0, f_1\}$. Since $f_0 \neq f_1$ there is $\bar{\eta} \in A$ such that $f_0(\bar{\eta}) \neq f_1(\bar{\eta})$.

Case 2. $\kappa \leq \text{cf}(\lambda)$, and if $\bar{\lambda} < \lambda$ and $\mu < \kappa$ then $\bar{\lambda}^\mu < \lambda$. If $|C(\eta)| < \lambda$ for all $\eta \in A$ we set $\bar{\lambda} = \sum_{\eta \in A} |C(\eta)|$ and we note that $\bar{\lambda} < \lambda$ (since $|A| = \mu < \kappa \leq \text{cf}(\lambda)$); hence we have

$$\lambda = |C| \leq \prod_{\eta \in A} |C(\eta)| \leq \bar{\lambda}^\mu < \lambda,$$

a contradiction.

The claim is proved.

We show next that $S_\lambda((X_J)_\kappa) > \gamma^\mu$. Since $|A| = \gamma$ and $\{U(f) : f \in \gamma^A\} \subset \mathcal{T}^*((X_J)_\kappa)$, it is enough to show that if $C \in [\gamma^A]^\lambda$ then $\bigcap_{f \in C} U(f) = \emptyset$. Let $C \in [\gamma^A]^\lambda$ and choose $\bar{\eta} \in A$ such that $|C(\bar{\eta})| = \lambda$. Since $\pi_{J(\bar{\eta})}[U(f)] = U^{f(\bar{\eta})}(\bar{\eta})$ for $f \in C$ we have

$$\bigcap_{f \in C} \pi_{J(\bar{\eta})}[U(f)] = \bigcap_{f \in C} U^{f(\bar{\eta})}(\bar{\eta}) = \bigcap_{\xi \in C(\bar{\eta})} U^\xi(\bar{\eta}) = \emptyset$$

and hence $\bigcap_{f \in C} U(f) = \emptyset$.

We note finally that

$$\alpha \geq S_\lambda((X_J)_\kappa) > \gamma^\mu.$$

as required.

The proof is complete.

We sharpen the estimates of Theorem 3.25 in the case $\kappa = \omega$.

3.27 Theorem. Let λ be a cardinal such that either $\lambda = 2$ or λ is (infinite and) regular, let $\{X_i : i \in I\}$ be a set of non-empty, Hausdorff spaces, and

$$\alpha = \sup \{S_\lambda(X_J) : \emptyset \neq J \subset I, |J| < \omega\}.$$

Then

$$\begin{aligned} S_\lambda(X_I) &= \alpha \text{ if } \alpha < \omega, \text{ or} \\ &\quad \alpha = \omega \text{ and } |X_I| < \omega, \text{ or} \\ &\quad \alpha > \omega \text{ and } \alpha \text{ is regular} \\ &= \alpha^+ \text{ in all other cases.} \end{aligned}$$

Proof. If $\alpha < \omega$ we have $S_\lambda(X_I) = \alpha$ from Theorem 3.25(b), and if $\alpha > \omega$ and α is regular we have $S_\lambda(X_I) = \alpha$ from Theorem 3.25(a). If $\alpha = \omega$ and $|X_I| < \omega$ then from Lemma 2.8 the relations $\lambda = 2$ and $\lambda > \omega$ both fail; hence $\lambda = \omega$ and from Lemma 2.8 we have $S_\lambda(X_I) = \omega = \alpha$. From Theorem 3.25(a) we have

$$\alpha \leq S_\lambda(X_I) \leq \alpha^+.$$

If $\alpha = \omega$ and $|X_I| \geq \omega$ then from Corollary 2.10 we have $S_\lambda(X_I) > \alpha$ and hence $S_\lambda(X_I) = \alpha^+$; and if $\alpha > \omega$ and α is singular, then $|X_I| \geq \omega$ and from Theorem 2.12 we have $S_\lambda(X_I) \neq \alpha$ and hence $S_\lambda(X_I) = \alpha^+$.

The proof is complete.

3.28 Theorem. If $\omega \leq \kappa < \alpha$ with α regular, then the following are equivalent.

(a) $\kappa \leq \alpha$;

(b) if $\{X_i : i \in I\}$ is a set of spaces with $d(X_i) < \alpha$ for $i \in I$, then $S((X_I)_\kappa) \leq \alpha$.

Proof. (a) $\Rightarrow$ (b). It follows from Theorem 3.25(a) (with β and λ replaced by α and 2, respectively) that it is sufficient to show that if $J \in \mathcal{P}_\kappa^*(I)$ then $S((X_J)_\kappa) \leq \alpha$. For $i \in J$ let D_i be dense in X_i with $|D_i| < \alpha$ and set $D = \prod_{i \in J} D_i$. It is clear that D is dense in $(X_I)_\kappa$. From Theorem A.5(c) we have

$$|D| = \prod_{i \in J} |D_i| < \alpha;$$

hence $S((X_J)_\kappa) \leq \alpha$.

(b) $\Rightarrow$ (a). If $\beta < \alpha$ and $\lambda < \kappa$ then the space $(\beta^\lambda)_\kappa$ is discrete, and from Lemma 2.8 we have $S((\beta^\lambda)_\kappa) = (\beta^\lambda)^+$. It follows from (b) that $(\beta^\lambda)^+ \leq \alpha$, i.e., $\beta^\lambda < \alpha$.

The P_κ -space $X_{(\kappa)}$ determined by a space X is defined in the Appendix.

3.29 Corollary. Let $\omega \leq \kappa \leq \alpha$ with κ and α regular and let $\{X_i : i \in I\}$ be a set of spaces such that $w(X_i) < \alpha$ for $i \in I$. Then $S((X_I)_{(\kappa)}) \leq \alpha$.

Proof. From the fact that κ is regular, it follows that if $\mathcal{B}$ is a base for X_i then $\{\cap \mathcal{A} : \mathcal{A} \in \mathcal{P}_\kappa(\mathcal{B})\}$ is a base for $(X_i)_{(\kappa)}$. Thus for

$i \in I$ we have

$$d((X_i)_{(i)}) \leq w((X_i)_{(i)}) \leq (w(X_i))^\kappa < \alpha,$$

and from Theorem 3.28 we have

$$S(\prod_{i \in I} (X_i)_{(i)}) \leq \alpha.$$

Since κ is regular the topology of $(\prod_{i \in I} (X_i)_{(i)})_\kappa$ is κ -complete; it is clearly the smallest κ -complete topology on the set X_I containing the product topology. Thus the spaces $(\prod_{i \in I} (X_i)_{(i)})_\kappa$ and $(X_I)_{(\kappa)}$ are equal and we have $S((X_I)_{(\kappa)}) \leq \alpha$, as required.

For every infinite cardinal λ there is by Lemma 2.8 a discrete space X such that $S_\lambda(X) = \lambda$. (For $\text{cf}(\lambda) = \omega$ and X a Hausdorff space, the relation $S_\lambda(X) = \lambda$ holds if and only if $0 < |X| < \omega$.) We note now that for λ (infinite and) regular every cardinal number not proscribed by the result of Chapters 2 and 3 does arise in practice as $S_\lambda(X)$ for an appropriate completely regular, Hausdorff space X .

3.30 Theorem. Let λ and α be cardinals such that $\lambda \leq \alpha$, α is uncountable and regular, and either $\lambda = 2$ or λ is (infinite and) regular. There is a completely regular, Hausdorff space X such that if I is a non-empty set then

(i) $S_\lambda(X^I) = \alpha$, and

(ii) every regular cardinal γ with $\gamma > \alpha$ is a calibre for X^I .

Proof. We show first there is X such that $S_\lambda(X) = \alpha$ and every regular cardinal γ with $\gamma \geq \alpha$ is a calibre for X . We consider two cases.

Case 1. α is a successor cardinal. Let $\alpha = \beta^+$ and let X be the (discrete) space β . Then $S_\lambda(X) = \alpha$ by Lemma 2.8, and X has calibre γ by Theorem 2.4(b) (with γ and α replaced by β and γ , respectively).

Case 2. α is a limit cardinal. We choose a set $\{\alpha_\sigma : \sigma < \alpha\}$ of cardinal numbers such that

$$\omega \leq \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } \sigma' < \sigma < \alpha,$$

$$\sum_{\sigma < \alpha} \alpha_\sigma = \alpha, \text{ and}$$

$$\lambda < \alpha_\sigma \text{ for } \sigma < \alpha \text{ if } \lambda < \alpha,$$

and we set $X = \prod_{\sigma < \alpha} \alpha_\sigma$. For $J \in \mathcal{P}_\omega^*(\alpha)$ we set $X_J = \prod_{\sigma \in J} \alpha_\sigma$ and we note that with $\sigma = \max J$ the space X_J is (homeomorphic to)

the discrete space α_σ . It follows from Theorem 3.27 and Lemma 2.8 that

$$\begin{aligned} S_\lambda(X) &= \sup \{S_\lambda(X_J) : J \in \mathcal{P}_\omega^*(I)\} = \sup \lambda = \lambda = \alpha && \text{if } \lambda = \alpha \\ &= \sup \{\alpha_\sigma^+ : \sigma < \alpha\} = \alpha && \text{if } \lambda < \alpha, \end{aligned}$$

as required. Further by Theorem 2.5(b) the (discrete) spaces α_σ have calibre γ ; hence X has calibre γ by Theorem 3.18(a).

The definition of X , and the proofs that $S_\lambda(X) = \alpha$ and X has calibre γ for every regular $\gamma \geq \alpha$, are complete in each case.

For $0 < n < \omega$ the space X^n is homeomorphic to X and hence $S_\lambda(X^n) = \alpha$. Thus

$$\sup \{S_\lambda(X_J) : J \in \mathcal{P}_\omega^*(I)\} = \alpha$$

and statement (i) follows from Theorem 3.27. Statement (ii) again follows from Theorem 3.18(a).

Notes for Chapter 3

The first mathematician to recognize that chain conditions in spaces X_i are reflected in (possibly weaker) properties of X_I was E. Marczewski ($\equiv$ E. Szpilrajn), who showed (Marczewski, 1941, Theorem 4(i)) that the product of second-countable spaces has calibre $(\omega^+, 2, 2)$ and (Marczewski, 1947, Theorem 3.3) that the class of spaces with calibre $(\omega^+, \omega^+, 2)$ is closed under the formation of Cartesian products. The proof of the former is self-contained and uses no principle of infinite combinatorics more complicated than the pigeon-hole principle of Dedekind and Dirichlet, but in the proof of the latter one can discern (p. 140) all ingredients necessary for (the appropriate case of) the Erdős–Rado theorem.

Marczewski (1941, 1(i)) noted Lemma 3.1 explicitly and he was apparently the first to raise the question (Marczewski, 1945, p. 304) whether the class of spaces with calibre $(\omega^+, 2, 2)$ is closed under the formation of Cartesian products.

Shanin proved (1946a) a version of the Erdős–Rado theorem for $\kappa = \omega$, he defined (1946b) the expression calibre α , and he showed (1946b, 1948) that for regular $\alpha > \omega$ the class of spaces with calibre α is closed under formation of Cartesian products.

We discussed Theorem 3.6(a, b) in connection with the abstract of

Comfort & Negrepontis (1978a); Theorem 3.6(c) is from Comfort & Negrepontis (1972b). Corollary 3.4 is a remark of Glicksberg (1959, p. 370), and (the case $Y = X$, of) Theorem 3.9 is due to Noble (1969b, Theorem 3.4).

Quasi-disjoint families were introduced (and named Δ -systems) by Erdős & Rado (1960); the notation $\alpha \rightarrow \Delta(\kappa, \beta)$ of Remark 3.8 is from Erdős & Rado (1969).

The device used in the proof of Theorem 3.10 to define a cover by pairwise disjoint open subsets of a product of discrete spaces has served related purposes in the past (see for example Mycielski, 1964; Engelking 1968, Exercise 3.2F(b); and Kemperman & Maharam, 1970). The κ -box product space $(X_i)_\kappa$ of Theorem 3.10, closely related to the example of Gerlits in Theorem 4.4 below, was suggested in correspondence by Eric K. van Douwen (in connection with Corollary 3.11 (iii)); see also Comfort (1979, Theorem 3.5).

In formulating the results of 3.13–3.15 we were motivated by Shelah (1977), who proved Lemma 3.17(a) and Theorem 3.18(a). Theorem 3.16 was communicated to us by Argyros (May, 1978), who found in addition a new proof (based on Corollary 1.11) of Shelah's Theorem 3.18(a); in Theorem 3.19 we have applied Argyros' basic argument to formulate Shelah-type theorems of considerable generality.

The case $\lambda = 2$ of Theorem 3.26(a) has been noted by Comfort & Negrepontis (1972a, Corollary 2.2) and (for $\kappa = \omega$) by Juhász (1971, Remark, p. 53). The results of 3.25 through 3.30 for $\lambda = 2$ are taken from Comfort & Negrepontis (1972a, 1974, 3.9–3.12); the case $\alpha = m^+$, $\kappa = n^+$, $m^n = m$ of Corollary 3.29 is due to Engelking & Karłowicz (1965, Theorem 6) (see also Engelking, 1965, Theorem 7), and the case $\lambda = 2$ of Theorem 3.30(i) was noted by Efimov (1968).

It is proved by Shanin (1946c, Theorem 5) that if α is a regular, uncountable cardinal and $d(X_i) < \alpha$ for $i \in I$, then $S(X_i) \leq \alpha$; this is the case $\kappa = \omega$ of Theorem 3.28. In Comfort & Negrepontis (1974, Remark, p. 78) we note that Theorem 3.28 can be proved, without any appeal to the Erdős–Rado theorem, by invoking the following (generalization of the) so-called Hewitt–Marczewski–Pondiczery theorem: if $\omega \leq \kappa \leq \alpha = \alpha^\kappa$ and $d(X_i) \leq \alpha$ for $i \in I$ with $|I| \leq 2^\alpha$, then

$d((X_i)_{i \in I}) \leq \alpha$. For a measure-theoretic proof (noted by Oxtoby, 1961) that every product X of separable spaces satisfies $S(X) \leq \omega^+$, see Theorem 6.7 and Corollary 6.8 below.

For an extensive treatment of cardinal numbers associated with topological spaces, including a number of results concerning calibres in product spaces and elsewhere, we refer the reader to Juhász (1971, 1980).

We note that from Corollary 3.14(a) it follows that if $\text{cf}(\alpha) > \omega$ and $\{X_i : i \in I\}$ is a set of spaces such that there is a set B of regular cardinal numbers satisfying

$$\beta < \alpha \quad \text{for } \beta \in B,$$

$$\sup B = \alpha, \text{ and}$$

$$X_i \text{ has calibre } \beta \text{ and calibre } \alpha \text{ for } \beta \in B, i \in I,$$

then X_I has calibre α . The same conclusion is given by Juhász (1980, Theorem 5.11) under the (stronger) hypotheses that $\text{cf}(\alpha) > \omega$ and $d(X_i) < \text{cf}(\alpha)$ for $i \in I$.